

PrefMem: A Preference-Aware, Evidence-Gated Agentic System for Interactive Robot Manipulation with Ambiguous Instructions

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Abstract

Vision–Language–Action (VLA) policies map language and visual observations to robot actions, but this interface does not by itself specify how an interactive robot should resolve under-specified intent, apply persistent user preferences, authorize an exact physical goal, recover from task-level failures, or establish that the complete goal has been achieved. This thesis presents PrefMem, a preference-aware, evidence-gated orchestration layer above an interchangeable action executor. PrefMem is not a new VLA and does not perform online policy learning. Five model-facing agents support human interaction, preference memory, planning, step monitoring, and final validation, while a deterministic runtime and receding-horizon controller own goal identity, confirmation, task publication, state transitions, temporal evidence requirements, and bounded recovery.

PrefMem is evaluated using development, non-confirmatory experiments in a restricted tabletop domain. With one applicable scoped preference, autonomous correct goal resolution increased from 0/36 observations under empty Memory to 36/36 under populated Memory, while both conditions achieved 36/36 final goal correctness after permitted clarification. Conflicting preferences caused clarification before commitment in 36/36 observations, and a held-out override replication selected the conflicting historical destination in 0/36 observations. A 30-interaction lifecycle experiment nevertheless retained the focal preference at the delayed final probe in only 2/8 stores, exposing a free-text identity limitation. Production Monitor improved the registered host route from 12/36 to 22/36 paired observations, while production Validator prevented false completion in all 21 visibly incomplete states compared with 0/21 for a stop-after-actions control. In twelve focused MuJoCo episodes, correct contracts and simulator-truth physical goals occurred in 12/12, all-strict grounding in 11/12, Validator completion in 9/12, and the exact one-action strict chain in 8/12.

These results support PrefMem as an inspectable systems architecture with measurable boundary-specific effects and identifiable failure mechanisms. They do not establish VLA superiority, general household competence, real-robot safety, or robust preference retention over unrestricted histories.

1 Introduction

1.1 Motivation

General-purpose robot policies increasingly combine visual observations, natural-language instructions, and continuous action generation. Brohan et al. (2022) demonstrated that a high-capacity transformer trained on large, diverse real-robot data can acquire scalable multi-task behaviour. Brohan et al. (2023) subsequently represented robot actions as tokens and co-fine-tuned a vision–language model on web and robot data, showing that semantic knowledge can transfer into control. Kim et al. (2024) introduced an openly available 7-billion-parameter VLA trained on 970,000 real-world robot demonstrations and reported an absolute 16.5-point advantage over RT-2-X across 29 tasks. The flow-matching action expert in π_0 targets dexterous continuous control across several embodiments (Black et al. 2025). These systems motivate the conceptual executor abstraction considered by this project: an instruction and an image are encoded into a multimodal representation, from which an action head predicts a continuous action or action chunk in a closed loop.

The direct vision-and-language-to-action formulation is powerful, but it does not by itself provide explicit mechanisms for several task-level decisions. A command can be syntactically clear and still be underdetermined for a particular person. “Tidy this up” does not say where each object belongs; “put the cube on my board” does not identify the board unless the robot can retrieve a scoped preference; and “do the usual thing” is unsafe when several incompatible histories are equally plausible. TidyBot showed that a language-model-based system can infer household placement preferences from a handful

of examples, achieving 91.2% accuracy on unseen benchmark objects and 85.0% placement success on its real robot (Wu et al. 2023). APRICOT further showed that preference inference must be reconciled with physical constraints and that active questions can disambiguate several plausible explanations (Wang et al. 2025). The present thesis starts from the same premise that personalisation matters, but focuses on how preference evidence should be scoped, confirmed, preserved, and carried into execution.

Long-horizon operation introduces a second source of risk. A multi-step plan generated from an old observation may become stale after the first action. A monitor may mistake transient contact for stable success, while a final image may be insufficient to prove every part of the original request. Research on grounded language planning has repeatedly shown the value of connecting semantic plans to affordances and feedback. SayCan scores language-model proposals with skill value functions (Ahn et al. 2022); Inner Monologue feeds success detection, scene descriptions, and human feedback back into planning (Huang et al. 2022); and ReAct interleaves reasoning with environment-facing actions and observations (Yao et al. 2023). More recent hierarchical systems place a reasoning model above a low-level VLA or policy, including Hi Robot and Agentic Robot (Shi et al. 2025; Yang et al. 2025b). PrefMem adopts the architectural lesson that semantic reasoning, physical execution, and verification should be separated, then adds explicit preference-memory and confirmation boundaries. The central problem is therefore not simply how to predict a motor command. It is how to maintain a trustworthy task contract between a person and an action-generating subsystem. The system must decide which historical preference is applicable, know when not to decide, preserve a current explicit correction over old memory, prevent a speculative preview from becoming an action, reject stale planning results, demand stable evidence after every action, and reserve final completion for a separate validation route. These responsibilities are difficult to audit when they are implicit in a single model prompt. PrefMem makes them explicit through typed messages and deterministic host state.

1.2 Research Problem and Aim

Existing work has separately studied preference inference, language-grounded planning, hierarchical execution, policy memory, and visual verification. This thesis studies the governance boundary connecting these concerns: how retrieved preference evidence becomes an explicitly authorised physical goal, and how that goal remains fixed while plans, actions, observations, and recovery decisions evolve.

Let a user issue instruction u in observed scene o_t , with user identity i , task scope s , and memory store $\mathcal{M}$. A useful system must form a confirmed goal contract g and then select actions a_t as observations evolve. The problem contains three interacting sources of uncertainty:

1. Semantic uncertainty: the instruction may omit an object, target, constraint, or preference.
2. Historical uncertainty: retrieved records may be irrelevant, stale, mutually inconsistent, or incorrectly scoped.
3. Execution uncertainty: grounding, control, observation, monitoring, and validation may disagree even when the intended goal is correct.

Accordingly, the aim of this thesis is to design and evaluate an inspectable orchestration layer that converts ambiguous, preference-conditioned user requests into confirmed physical-goal contracts and governs their execution through fresh planning, temporally fenced evidence, bounded recovery, and independent final validation. The thesis addresses the following research questions:

1. To what extent does one applicable user- and task-scoped preference reduce clarification requirements while preserving final goal correctness?
2. Under what memory conditions does PrefMem abstain from autonomous commitment, and how reliably does it preserve explicit current instructions over historical defaults?
3. How reliably does the implemented preference lifecycle preserve user, task, record, and value identity through creation, retrieval, update, intervening mutations, and revocation?
4. What boundary-specific effects do production Monitor and Validator have relative to their registered controls?
5. To what extent can a confirmed preference-conditioned contract reach its registered MuJoCo final state, and which component boundaries explain unsuccessful strict completion?

1.3 Contribution

This thesis has the following contributions.

1. A five-agent orchestration architecture separating human interaction, preference memory, planning, step monitoring, and final validation from an interchangeable physical executor.
2. A deterministic trust boundary in which models propose semantic content while host code owns goal identity and revision, exact confirmation, lifecycle state, an execution horizon of one, publication fencing, temporal evidence aggregation, loop guards, and terminal state.
3. A conservative preference-memory policy combining retrieval before mutation, bounded candidate injection, semantic applicability filtering, explicit future-facing consent, and at most one justified mutation per interaction.
4. A confirmed execution contract consisting of an immutable goal and frozen validation checklist, with step-level monitoring, fresh replanning after terminal results, and a separate whole-goal completion gate.
5. A layered evaluation methodology that reports direct component, interface, ablation, robustness, and end-to-end outcomes while preserving null and negative evidence and avoiding a misleading global system score.

2 Related Work

PrefMem lies at the intersection of language-conditioned robot control, personalised assistance, interactive task specification, long-horizon orchestration, and visual verification. These areas address different responsibilities. A robot policy determines how an instruction should be realised through actions, whereas an orchestration system must additionally determine whether the instruction is sufficiently specified, which historical information is authorised and applicable, what exact outcome the user has confirmed, and what evidence permits the system to advance or terminate. This section therefore distinguishes action-generation capability from task-level governance and positions PrefMem as a complementary layer above an interchangeable executor.

2.1 Language-Conditioned Robot Policies and VLAs

Large-scale language-conditioned robot policies established that diverse robot data and high-capacity architectures can support multi-task behaviour. RT-1 introduced a transformer-based policy trained on a large collection of real-robot trajectories and studied how performance varied with model capacity, dataset size, and task diversity (Brohan et al. 2022). RT-2 subsequently formulated robot actions as text tokens and co-fine-tuned vision-language models on robot trajectories and Internet-scale vision-language tasks (Brohan et al. 2023). This allowed semantic knowledge acquired during vision-language pretraining to influence action generation, including generalisation to previously unseen object categories, linguistic references, and elementary reasoning tasks.

Related work has explored embodied reasoning without directly treating the model as a complete action policy. PaLM-E injects continuous sensor representations into a large language model and jointly trains across embodied reasoning, visual question answering, and language tasks (Driess et al. 2023). It demonstrates that multimodal foundation models can integrate physical observations with semantic reasoning, but does not imply that the same model must own task authorisation, persistent user preferences, or the complete robot lifecycle.

OpenVLA provides an open 7-billion-parameter VLA based on a Llama 2 language backbone and fused DINOv2 and SigLIP visual features (Kim et al. 2024). It was trained on approximately 970,000 real-world robot demonstrations and demonstrated efficient adaptation to new tasks. Continuous-action architectures have developed alongside token-based methods. The π_0 model combines a pretrained vision-language backbone with a flow-matching action expert and trains across single-arm, dual-arm, and mobile manipulator embodiments (Black et al. 2025). This formulation supports continuous action chunks for dexterous behaviours while retaining semantic conditioning from the vision-language backbone.

More recent systems increase the scale and steerability of the policy itself. LingBot-VLA reports pretraining on approximately 20,000 hours of real-world data from nine dual-arm configurations and evaluates transfer across multiple robot platforms (Wu et al. 2026). The later $\pi_{0.7}$ model conditions

action generation on a richer multimodal context, including subtask instructions, generated subgoal images, control mode, and episode-quality metadata (Physical Intelligence et al. 2026). This detailed conditioning allows data containing different strategies, qualities, and embodiments to be used without requiring the policy to average them into a single undifferentiated behaviour.

These advances make it inappropriate to motivate PrefMem through a categorical claim that VLAs cannot reason or generalise. Modern VLAs increasingly support language understanding, continuous control, memory, and multi-stage behaviour. The remaining problem addressed in this thesis is different: a capable action policy does not by itself define how persistent preference evidence becomes an authorised goal, how a user confirms that goal, or how task-level evidence is fenced and audited across execution.

2.2 Semantic Reasoning Within VLAs

A growing body of work studies how the semantic capabilities of a pretrained vision-language model can be preserved or strengthened during action training. One challenge is that optimisation on robot trajectories may cause catastrophic forgetting or encourage the backbone to correlate familiar visual and linguistic patterns directly with actions. This can reduce the benefit of the pretrained model’s broader reasoning capabilities.

InstructVLA addresses this problem through Vision-Language-Action Instruction Tuning, a two-stage training procedure that combines multimodal reasoning data, embodied instruction annotations, mixture-of-experts adaptation, and flow-based action generation (Yang et al. 2025a). Its instruction-tuning data include scene descriptions, question answering, command rewriting, and implicit task contexts. Consequently, the model is trained not only to execute atomic instructions but also to interpret paraphrases, attributes, negation, and higher-level task descriptions.

SemanticVLA similarly studies the tension between semantic reasoning and action-pattern memorisation (Ni et al. 2026). It introduces two complementary intermediate representations. Explicit trace reasoning generates interpretable spatial waypoints through the language interface, while latent action tokens encode compact, visually grounded manipulation primitives. Both signals condition a flow-matching action decoder. Its instruction-rephrasing experiments are particularly relevant because they evaluate whether semantically equivalent instructions continue to produce appropriate actions rather than merely matching wording observed during training.

These methods improve reasoning *inside* the execution policy. PrefMem instead externalises several task-level decisions. Its preference provenance, goal confirmation, publication identities, execution history, and final completion report remain inspectable even when the executor uses opaque latent representations. The approaches are therefore complementary: an executor such as InstructVLA, SemanticVLA, or $\pi_{0.7}$ could replace PrefMem’s current execution surrogate without changing the upper-level goal, evidence, or control contracts.

2.3 Personalised Robot Assistance and Preference Learning

Personalised assistance requires a robot to distinguish a user’s habitual choices from generic common-sense. TidyBot studies this problem in household cleanup, where the correct receptacle and manipulation primitive depend on individual preferences (Wu et al. 2023). The system uses an LLM to summarise a small set of example placements into general rules and applies those rules to unseen objects. This demonstrates that language models can generalise personalised placement patterns from limited examples. However, its principal research question is preference generalisation, rather than the lifecycle and authority of individual persistent records.

LLM-Personalize adapts an LLM planner to household preferences through imitation learning and reinforced self-training (Han et al. 2025). The planner acts iteratively in partially observable House-keep environments using a dynamically constructed scene graph. An initial imitation-learning phase teaches executable planning and preliminary preference alignment, after which self-training explores and reinforces preference-compatible plans. In this case, personalisation is incorporated through model optimisation. This differs from PrefMem, which keeps the underlying conversational and action models fixed and represents preferences as explicit, inspectable records.

APRICOT combines LLM-based Bayesian active preference learning with constraint-aware task planning (Wang et al. 2025). Visual demonstrations are converted into language descriptions and used to construct candidate preference hypotheses. When uncertainty remains, the system selects questions intended to reduce the posterior uncertainty over those candidates. A task planner then refines its proposed arrangement using geometric feedback from a world model. APRICOT therefore connects

preference inference, active questioning, and physical feasibility. Its active-learning formulation assumes that an adequate explanation is represented among the candidate preferences, whereas PrefMem treats missing or conflicting applicable records as reasons to abstain and request direct clarification.

PrefMem does not attempt to infer a general reward function or retrain a personalised planner from demonstrations. It instead addresses persistent preference governance: records are retrieved under user and task scope, filtered for semantic applicability, mutated only through an explicit memory operation, and never created merely because the user confirmed the current task. A current instruction has field-level priority over historical defaults. Thus, the central distinction is between *learning what a user probably prefers* and *deciding whether an existing preference record is authorised, applicable, and sufficient for the present physical task*.

2.4 Ambiguity, Clarification, and Human Assistance

A robot should not resolve every uncertain instruction by guessing, but asking for help on every task would make the system unnecessarily dependent on the user. KnowNo formalises this trade-off as uncertainty alignment for language-model planners (Ren et al. 2023). It uses conformal prediction to construct a calibrated set of candidate actions. A singleton set allows autonomous execution, whereas a larger set causes the robot to ask the user for help. Under its calibration assumptions, this provides a user-specified coverage level while reducing unnecessary intervention.

KnowNo is an important comparison because it provides a formal uncertainty guarantee, which PrefMem does not. Its guarantee assumes, among other conditions, that the relevant environment entities are grounded in the planner’s input and that the proposed low-level actions can be executed successfully. PrefMem instead separates semantic, historical, and execution uncertainty. It may clarify because a task field is missing, because retrieved records conflict, because the Planner reports a capability boundary, or because monitoring and validation cannot obtain sufficient evidence. Clarification and attention are governed by typed system states and semantic policy rather than a calibrated conformal prediction set.

APRICOT and KnowNo both show that question asking should be selective rather than universal. PrefMem follows the same interaction principle at a narrower preference-memory boundary. One uniquely applicable record may provide a default for a missing field; an empty, irrelevant, or conflicting result requires a focused question. Regardless of whether clarification was required, PrefMem subsequently asks the user to confirm the exact staged goal identifier and revision. This confirmation is an authorisation boundary and should not be confused with statistical uncertainty calibration.

2.5 Language Planning and Agentic Robot Orchestration

Language-model planning systems often combine semantic proposals with embodiment-specific constraints. SayCan scores language-model skill proposals using value functions associated with executable robot skills (Ahn et al. 2022). The language model contributes semantic knowledge about how a task may be decomposed, while the value functions ground this knowledge in the robot’s available capabilities and current environment. Code as Policies instead prompts a language model to generate programs that compose perception and control APIs (Liang et al. 2022). These programs can express geometric reasoning, feedback loops, and reusable hierarchical functions.

Closed-loop feedback is necessary because an initially plausible plan may become invalid during execution. Inner Monologue feeds scene descriptions, success-detector outputs, and human feedback back into a language planner (Huang et al. 2022). This enables the planner to revise its decisions as execution changes the environment. ReAct provides a broader language-agent formulation in which reasoning traces are interleaved with environment-facing actions and observations (Yao et al. 2023). Although ReAct is not specifically a robot-control architecture, its reasoning-acting loop influenced later agentic systems.

Hi Robot explicitly divides robot control into a high-level VLM and a low-level VLA (Shi et al. 2025). The high-level model interprets complex user prompts, visual observations, and mid-task feedback, then produces simpler instructions for the low-level policy. Its evaluation demonstrates the value of hierarchy for situated corrections and compositional instructions. The high-level policy is invoked at a lower frequency or in response to user feedback, while the low-level policy continuously produces action chunks.

Agentic Robot introduces a Planner–Executor–Verifier architecture governed by a Standardized Action Procedure (Yang et al. 2025b). A reasoning model decomposes a task into atomic subgoals, a VLA executes each subgoal, and a temporal VLM verifier determines whether execution should continue, retry,

or recover. This work is a close architectural precedent for PrefMem because it treats verification as a control signal rather than an offline metric.

RoboClaw extends agentic orchestration across the robot learning lifecycle (Li et al. 2026). A VLM meta-controller reasons over observations, task memory, working memory, and a tool interface to select and monitor learned skill policies. Forward and inverse policies form self-resetting data-collection loops, while deployment trajectories can later refine the policy library. RoboClaw therefore connects orchestration with continued data acquisition and policy improvement.

PrefMem adopts the general lesson that semantic planning, execution, and verification benefit from explicit interfaces, but introduces different authority and temporal semantics. The Planner may predict several candidate tasks, yet deterministic host code publishes only the first, producing an execution horizon of one. After each stable terminal step result, a new frame and immutable execution history are used to request a fresh plan toward the same frozen goal. In contrast to RoboClaw, PrefMem does not update policy weights from deployment experience. In contrast to systems that rely only on an initially decomposed subgoal sequence, PrefMem discards the unexecuted prediction tail and replans after every terminal task observation.

2.6 Memory Across Robot Interaction

The term *memory* describes several distinct mechanisms in embodied systems. A policy may require short-term visual memory to handle occlusion, within-episode semantic memory to track completed subtasks, or persistent declarative memory describing a user’s preferences across separate interactions. Treating these mechanisms as interchangeable obscures both their authority and their failure modes.

Multi-Scale Embodied Memory (MEM) combines a video-based short-horizon memory with a compressed language-based long-horizon memory inside a VLA (Torne et al. 2026). The video encoder retains fine-grained recent information needed for dynamics, re-grasping, and self-occlusion. A high-level policy maintains a textual summary of relevant events, such as which recipe steps have been completed or which doors remain open. This permits the policy to act over tasks lasting many minutes without passing the complete dense observation history into every inference call.

RoboClaw similarly distinguishes task-level and working memory. Its task memory records global progress, whereas working memory records the currently active skill and tool history (Li et al. 2026). These forms of memory support execution context. TidyBot and APRICOT, by comparison, represent information about a user’s desired organisation rules (Wu et al. 2023; Wang et al. 2025).

PrefMem contains both task history and preference memory, but keeps them under different contracts. Terminal execution records are host-owned, immutable evidence supplied to future planning cycles. Persistent preference records are stored separately and may be retrieved, remembered, updated, or forgotten only through the Memory boundary. Conversation history is evidence for a question; it is not automatically converted into a persistent default. This separation is central to the thesis because a record of what happened during one task does not, by itself, authorise the robot to treat that event as the user’s future preference.

2.7 Monitoring, Verification, and Goal Completion

Execution feedback can serve several purposes. A success detector may indicate that the next subgoal can begin, a verifier may identify a failed grasp and trigger recovery, and a task-level evaluator may determine whether the user’s complete request has been satisfied. Prior closed-loop systems primarily emphasise the first two responsibilities. Inner Monologue incorporates success detections into replanning (Huang et al. 2022); Agentic Robot uses a temporal verifier to decide whether to continue, retry, or recover (Yang et al. 2025b); and RoboClaw periodically queries robot and environment state to supervise skill execution (Li et al. 2026).

These mechanisms address the limitations of open-loop progression, but local subgoal success is not logically equivalent to whole-goal completion. A robot may successfully place one object while violating a required order, disturbing another completed arrangement, or leaving part of the original request unfinished. Conversely, an occluded or ambiguous view should not automatically be interpreted as physical failure.

PrefMem therefore separates two visual authorities. Monitor evaluates only the single active step and its expected observations using post-publication evidence. Validator evaluates an immutable checklist derived from the complete confirmed goal. The checklist is frozen before execution, preventing later failures from weakening the success criteria. Validator may accumulate criterion evidence over multiple

views, while deterministic host code derives whether the outcome is complete, incomplete, or still requires evidence. This separation is intended to prevent attempted actions or a count of successful subtasks from being treated as proof of overall completion.

2.8 Positioning of PrefMem

Table E.1 summarises the closest relationships. The categories are architectural rather than claims of strict feature absence: several systems could potentially be extended with additional memory or verification components. The comparison identifies the primary responsibility evaluated by each work and the boundary that distinguishes PrefMem.

The complete comparison is retained in Appendix Table E.1.

Taken together, prior work demonstrates increasingly capable robot policies, effective preference inference, selective help-seeking, hierarchical reasoning, task memory, and visual verification. The research gap addressed by PrefMem is not the absence of any one of these capabilities. It is the inspectable conversion of scoped preference evidence into an authorised physical-goal contract, followed by deterministic governance of task publication, evidence acceptance, bounded recovery, and terminal completion. PrefMem consequently contributes a systems and evaluation architecture rather than a new action model: the executor may improve independently while the preference, authority, and evidence boundaries remain explicit.

3 Methodology

This chapter derives PrefMem’s architecture from the research problem rather than treating the implemented agents as an arbitrary decomposition. It first formalises the task-level control problem and introduces Graph Engineering and Loop Engineering as operational design lenses. It then evaluates the simpler alternative of one tool-using agent, explains why edge deployment, context isolation, authority, temporal coordination, and verification requirements motivate role specialisation, and presents the selected architecture. The remainder of the chapter describes the deterministic control plane, each model-facing agent, the execution adapter, their typed interactions, deployment model, recovery boundaries, and requirement-to-evidence traceability. Graph and Loop Engineering organise the methodology; they are not claimed as new formal disciplines or as empirical results of this thesis.

3.1 Problem Setting and Assumptions

At task-level cycle k , PrefMem receives a user instruction u , a current visual observation o_k , user identity i , task scope s , and a persistent preference store $\mathcal{M}$. The system must first construct an exact goal contract g that the user authorises. It must then publish a sequence of task-level instructions $z_1, z_2, \dots$, each of which is executed by a lower-level interface that produces physical actions a_t . The task-level system observes the resulting scene, records trusted terminal evidence, and either plans the next task, requests more evidence, pauses for human attention, or declares the frozen goal complete.

The resulting closed-loop orchestration problem is summarised by

$$\begin{aligned} \tilde{g} &= \Gamma(u, i, s, \mathcal{M}, o_0), & g &= \tilde{g} \quad \text{iff} \quad C(\text{id}(\tilde{g}), \text{rev}(\tilde{g})) = 1, \\ d_k &= \Pi(g, o_k, \mathcal{H}_{k-1}, r_k), & z_k &= \text{Select}_1(d_k), \\ e_k &= \Omega_{\phi_k}(z_k, \{o_t \mid t > \tau_k\}), & x_{k+1} &= \delta(x_k, d_k, e_k), \\ x_K &= \text{COMPLETE} \iff V(g, \mathcal{E}_K) = \text{COMPLETE}. \end{aligned} \tag{1}$$

Here, Γ constructs the scene-grounded proposal $\tilde{g}$, and C authorises it only when the user confirms its exact identifier and revision. Planner Π receives the immutable confirmed goal, current observation, trusted terminal history $\mathcal{H}_{k-1}$, and trigger or operator guidance r_k . For an action decision, Select_1 publishes only the first candidate task. Ω_{ϕ_k} denotes the phase-dependent evidence process—Monitor for a normal step and Validator for final validation—using only observations newer than publication time τ_k . The deterministic transition function δ updates authoritative controller state x_k . The operator V denotes deterministic host aggregation of the accumulated Validator evidence $\mathcal{E}_K$; terminal completion is permitted only when that aggregation establishes the frozen goal as complete.

The formulation rests on six assumptions. First, the user remains the source of authority for ambiguous task semantics and persistent preference changes. Second, camera observations are partial and may be stale, occluded, or insufficient for a terminal claim. Third, every model output is fallible and is

therefore a proposal or assessment rather than authoritative state. Fourth, models do not assign trusted identifiers, append execution history, or perform controller transitions. Fifth, model weights remain fixed during a PrefMem session; the system is not an online policy-learning method. Finally, the physical executor is replaceable. The current MuJoCo path combines a constrained language compiler, SAM-based grounding, RGB-D geometry, and deterministic control. It is a VLA-shaped execution surrogate rather than a learned VLA contribution.

3.2 Design Method: Graph Engineering and Loop Engineering

Graph Engineering. In this thesis, *Graph Engineering* means explicitly designing an agentic system as typed state, specialised processing nodes, and guarded edges. This usage follows graph-based agent frameworks in which state records the current application snapshot, nodes perform model or deterministic work, and edges select the next node through fixed or conditional transitions (LangChain 2026). The important object is therefore not a visual diagram but an executable contract: who owns each state field, which node may update it, what data may traverse an edge, and which predicates make a transition legal.

PrefMem’s problem is naturally graph-shaped. Clarification, memory retrieval, goal preview, confirmation, planning, execution, monitoring, validation, and recovery have different inputs and authority. A physical publication must be unreachable before exact confirmation; a Memory mutation must be unreachable before request-scoped retrieval; a Monitor result must not be routed into a final-validation transition; and terminal completion must be unreachable without the frozen checklist. Representing these constraints as graph edges makes invalid paths structurally rejectable rather than merely discouraged by prompt wording.

Loop Engineering. *Loop Engineering* is used operationally for the design of recurrent workflows that pursue a goal, act, observe, evaluate progress, adjust, and stop against explicit evidence (Belcic and Stryker 2026). Agent-engineering guidance similarly emphasises environmental feedback, checkpoints, stopping conditions, and iteration guards (Anthropic 2024). A loop is not reliable simply because a model can call itself again. Its state, progress signal, evidence freshness, failure route, and maximum iteration policy must be engineered.

PrefMem contains nested loops at different temporal scales. The interaction loop alternates among user input, HRI reasoning, and optional Memory calls until the goal is sufficiently specified. The Memory mutation loop requires retrieval before at most one write. The execution loop alternates among fresh planning, one-task publication, physical execution, post-publication observation, and terminal history. The validation loop accumulates views until the frozen checklist is complete, incomplete, or still lacks evidence. Recovery loops can republish a current task or request a new plan, while cycle, failure, repetition, and timeout guards bound every automatic path.

Graph Engineering and Loop Engineering consequently answer complementary questions. Graph Engineering determines who may act and which transitions are legal; Loop Engineering determines how cyclic paths demonstrate progress and terminate. The deterministic controller connects the two by converting typed events into guarded state-machine transitions.

3.3 Architectural Alternatives and Design Decision

The simplest agentic implementation would expose every prompt, tool, memory, runtime control, and validation function to one conversational agent. Such a monolithic agent could retrieve a preference, construct a plan, execute tools, observe the result, and answer the user in one history. It would require fewer model calls, reduce orchestration code, and potentially lower latency. PrefMem does not assume that this design is incapable; it rejects it because it is a poor match for the project’s resource, authority, timing, and auditability requirements.

Context and tool pollution. A monolithic prompt would combine user dialogue, preference candidates, planning history, active-step criteria, final-state criteria, runtime state, and every available tool schema. Irrelevant context can materially reduce LLM reasoning accuracy (Shi et al. 2023), while evidence placed inside a long context is not used uniformly (Liu et al. 2023). RULER further shows that advertised context length is not equivalent to robust multi-hop use of that context (Hsieh et al. 2024). Function-calling benchmarks likewise report degradation for some models as conversations and available API sets grow, with chained calls remaining difficult (Alkhoul et al. 2025). These findings do not prove

that PrefMem’s decomposition outperforms a monolith, because that ablation has not been run; they motivate context minimisation as an engineering precaution.

PrefMem therefore exposes only task-relevant information. HRI receives the conversation, current frame, authoritative runtime summary, and five safe intent-level tools. Memory receives one labelled request and a capped candidate set. Planner receives the frozen goal, one fresh frame, trigger, and trusted terminal history. Monitor receives one publication and its step criteria. Validator receives the frozen checklist and final-state views. Role separation reduces prompt burden without duplicating the foundation-model weights.

Authority and error containment. The monolithic agent would be able to propose a preference, persist it, stage a goal, publish an action, and judge its own result. Separating these capabilities follows modular decomposition and information hiding (Parnas 1972), as well as least privilege (Saltzer and Schroeder 1975). HRI cannot append history, Planner cannot write Memory, Monitor cannot publish a task, Validator cannot rewrite the goal, and no model can assign trusted identifiers or enter **COMPLETE**. This limits the effect of a malformed or semantically incorrect call to a typed boundary.

Separating planning from validation also reduces direct self-certification. LLM evaluators can exhibit self-preference toward their own generations (Panickssery et al. 2024). PrefMem’s logical separation does not provide full statistical independence because the roles currently share the Gemma backbone; it instead provides prompt, context, evidence, and authority independence. Correlated model errors remain a limitation and motivate a future heterogeneous or geometry-assisted Validator.

Temporal separation and testability. The roles operate at incompatible rates. HRI runs on user turns, Memory only on explicit requests, Planner once per task-level cycle, Monitor repeatedly during one publication, Validator during compilation and final validation, and the executor at a much higher control frequency. A single conversational history would need to multiplex these clocks and decide whether delayed outputs still belong to the active task. Separate services and publication identities make stale-event rejection explicit. They also permit isolated component tests, boundary-specific ablations, and failure attribution to retrieval, dialogue, planning, monitoring, validation, grounding, or control.

The complete monolithic-versus-role-specialised trade-off is summarised in Appendix Table E.2.

The selected design is thus a *single-model, multi-role* architecture: the five agents are logically distinct but share the primary VLM endpoint. This preserves local deployment feasibility while externalising context, authority, and temporal structure. The choice is a requirements-driven design decision, not a demonstrated universal superiority claim over monolithic agents.

3.4 Design Requirements

The research problem yields the following system requirements:

User authority: A materially ambiguous physical outcome must be clarified or resolved from uniquely applicable authorised evidence before an executable goal is formed.

Preference provenance: Persistent records must retain identity and scope, and a current explicit instruction must take priority over a historical default for the same field.

Exact confirmation: Physical execution must require confirmation of the exact staged goal identifier and revision.

Fresh horizon-one planning: Only one task may be published per cycle, and every subsequent cycle must use fresh scene evidence and immutable terminal history.

Evidence fencing: Assessments must match the active goal, revision, publication, phase, and temporal evidence window before they may affect controller state.

Stable terminal decisions: A single transient observation must not be sufficient to establish task success or failure.

Independent completion: Local action or subtask success must not be treated as proof that the complete user goal is satisfied.

Bounded recovery: Replanning, retries, and timeout recovery must be guarded so that repeated failure produces an inspectable attention state rather than an unbounded loop.

Executor independence: The task-level contracts must remain valid when the human or MuJoCo execution adapter is replaced by a future VLA.

R10: Inspectability: Goal state, publication identity, terminal history, accepted evidence, recovery causes, and completion criteria must be available for audit and experiment scoring.

3.5 System Architecture

Figure 1 presents the selected architecture at component level. The overview deliberately collapses the internal agent loops: each dense component is expanded in a separate role-specific subgraph later in this section. Five model-facing roles share the configured conversational VLM while receiving different prompts, state, images, tools, and schemas. HRI is the sole user-facing role. Memory owns preference retrieval and model-mediated mutation. Planner provides a non-executable preview and fresh task-level decisions. Monitor observes one active step. Validator compiles and assesses the frozen whole-goal checklist. Logical separation limits the context and authority of each call; it does not imply five copies of the model in GPU memory.

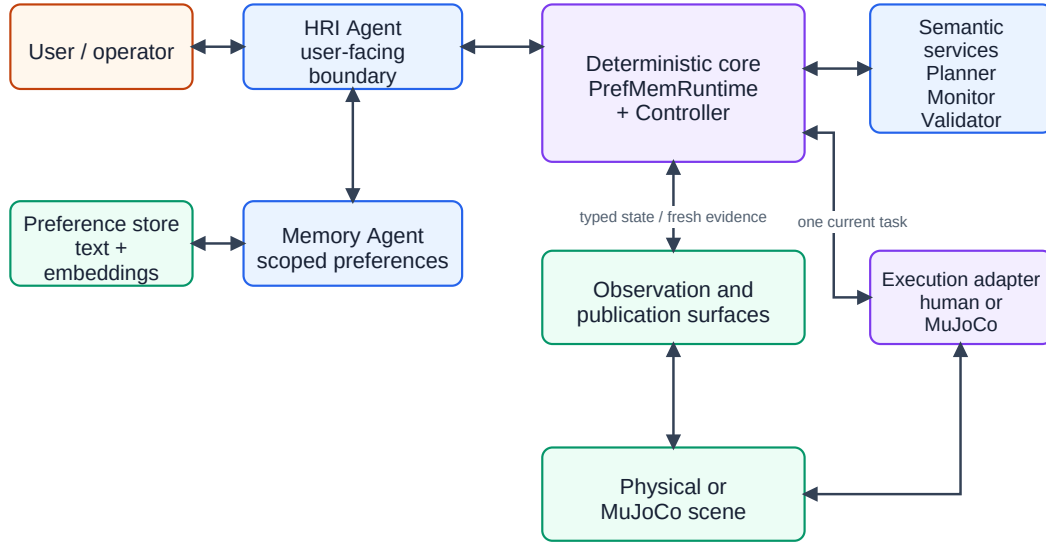


Figure 1: Overall PrefMem component architecture. The semantic-services and execution-adapter blocks are expanded in the role-specific workflow subgraphs that follow. Blue nodes are model-facing roles, purple nodes are deterministic host components, green nodes are state or evidence surfaces, and orange nodes are human roles.

Two deterministic components form the control plane. **PrefMemRuntime** owns model and service composition, frame capture, task display publication, observation-service routing, notifications, executor integration, and the shared emergency coordinator. **RecedingHorizonController** owns the frozen goal, cycle number, current publication, terminal history, evidence streaks, attention reason, and loop guards. It performs no model calls and no I/O; instead it returns explicit transitions for Runtime to execute.

The executor is connected through one **PublishedTask**. In human mode, the current instruction is displayed through the camera-page task API. In MuJoCo mode, the same publication is sent to **ExecutionService**. The simulation atomically produces two synchronised observation streams. The robot-hidden **sam** RGB-D view is reserved for grounding and execution, whereas the third-person **prefmem** view preserves the visible robot for HRI, Planner, Memory, Monitor, and Validator. This separation prevents semantic agents from receiving simulator truth while allowing the executor to use calibrated depth.

Persistent preference text is stored in **preference.json**, with normalised EmbeddingGemma vectors in **preference.npy**. Terminal task history is not stored in this preference database: it is immutable

controller-owned evidence scoped to one confirmed goal. The task publication surface is likewise a single current slot rather than a queue, ensuring that a human or robot sees only the authoritative active instruction.

3.6 Deterministic Control Plane and Typed Contracts

Figure 2 gives the coarse controller state graph. To keep the main graph legible, the active receding-horizon loop and its exception handling are represented as components and expanded separately in Figures 3 and 4. A proposed goal moves the controller from **IDLE** to **AWAITING_CONFIRMATION**. Exact confirmation starts the active loop. Planner can route to one executable step, final validation, a blocker, or a user question. Stable terminal evidence returns execution or incomplete validation to planning, while only stable whole-goal validation can enter **COMPLETE**.

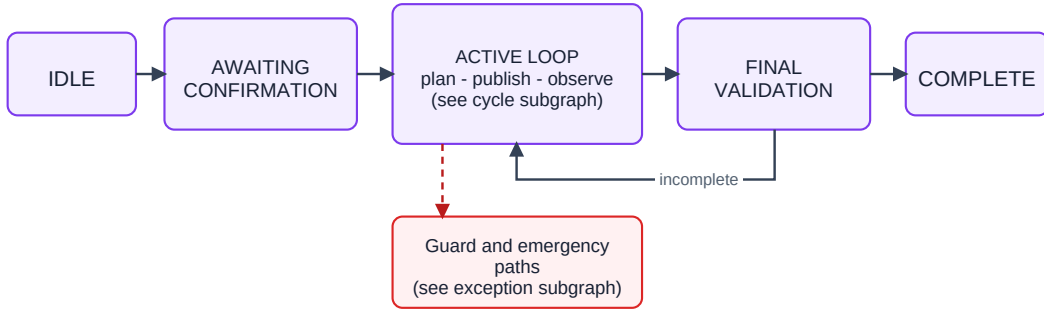


Figure 2: Top-level controller states. The active loop and guard paths are shown as components here and expanded in Figures 3 and 4.

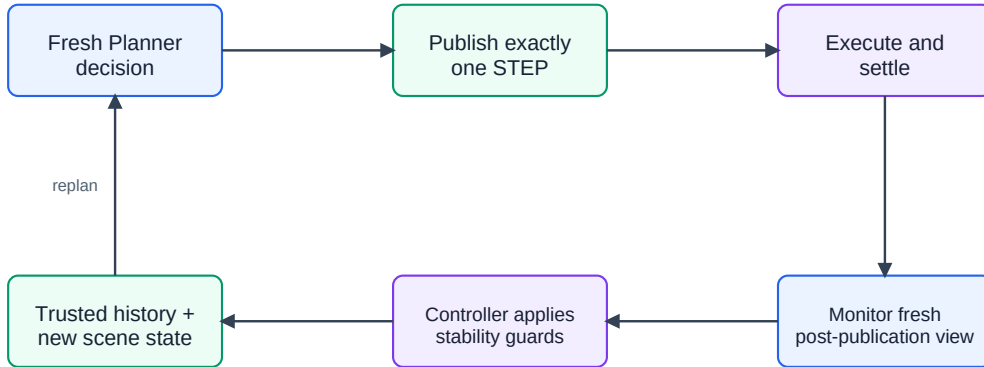


Figure 3: Expanded active receding-horizon loop. Only one step is published; fresh post-publication evidence is converted into trusted history before the next Planner call.

Typed contracts make graph edges executable and auditable. A **GoalProposal** contains the scene-grounded proposed outcome, explicit constraints, nominal outline, and observable final state. Runtime converts a non-blocked proposal into a **GoalContract** with a trusted goal identifier and revision. A **ValidationContract** maps each broad final outcome to detailed visual criteria and is frozen at confirmation. A **PlannerCycleRequest** binds the immutable goal to a fresh frame, trigger, trusted terminal history, and optional operator guidance.

For an action decision, host code selects only `candidate_tasks[0]` and constructs a **PublishedTask** with trusted cycle, step, criterion, publication, phase, and timestamp fields. Monitor and Validator responses are parsed into typed assessments and rejected if their identifiers, ordered criterion set, frame sequence, or phase do not match authoritative state. Only deterministic transitions append an

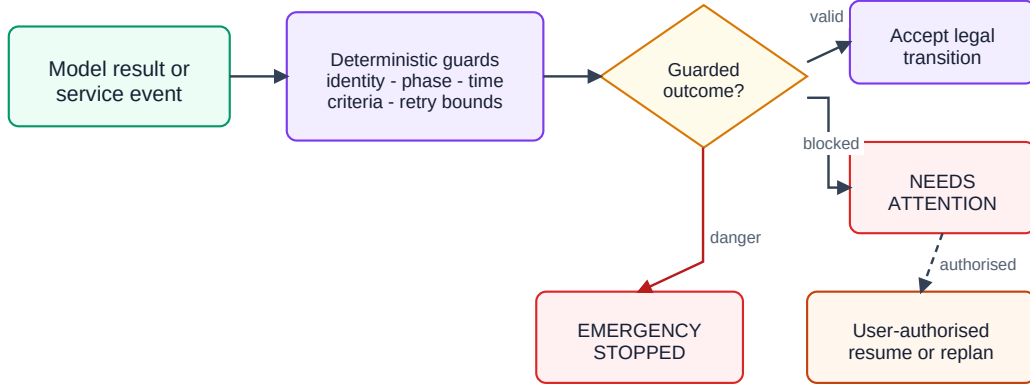


Figure 4: Expanded controller guard and exception routing. Invalid or exhausted operations enter `NEEDS_ATTENTION`; valid danger evidence enters the process-terminal `EMERGENCY_STOPPED` state.

`ExecutionRecord`, request another planning cycle, enter attention, or establish completion. This implements the rule that models may propose semantics, but host code owns lifecycle state.

3.7 HRI Agent Design

Responsibility and inputs. HRI is the only role permitted to converse with the user. Each model call receives the current user message, a current `prefmem` frame, conversation messages, and an injected authoritative runtime summary. That summary exposes the staged or confirmed goal, current publication, terminal history, validation report, attention cause, and currently authorised recovery actions. Conversation history helps interpret the current turn but is not authoritative state and does not by itself authorise a new preference.

Tool surface and authority. Once Runtime is attached, the generic sub-agent tool is replaced by five strictly typed tools: call Memory, request a goal preview, confirm an exact goal revision, resume the current publication, or request replanning with optional guidance. Planner cycles, publication, Monitor/Validator control, history mutation, and emergency reset are deliberately absent. Tool errors are returned as structured observations so the dialogue can report them without bypassing the control plane; a latched emergency disables further tool execution.

Workflow. Figures 5 and 6 separate the HRI loop into intent resolution and contract confirmation. HRI first determines whether the physical outcome and important constraints are sufficiently clear. For an under-specified reference, it requests Memory before guessing. One applicable non-conflicting preference may fill the missing field; empty, irrelevant, or conflicting evidence produces one focused clarification.

Once the request is clear, HRI asks Runtime for a scene-grounded preview and presents the exact goal, constraints, expected outcomes, nominal strategy, goal identifier, and revision. It may invoke confirmation only after the user refers to that exact identity. Rejection retains the proposal for discussion but authorises no execution.

HRI also translates controller notifications into a user-facing report. On completion it exposes the broad frozen checklist rather than internal detailed criteria; on attention it reports the authoritative cause and only the recovery actions currently legal. The HRI/Memory/contract boundary is evaluated by the HRI component and interface studies and by focused EXP-01 through EXP-05. These experiments test the combined decision boundary rather than proving universal natural-language understanding.

3.8 Memory Agent Design

Store and request boundary. Memory maintains a list of preference records and a row-aligned matrix of normalised embeddings. Requests must begin with `RETRIEVE REQUEST:` or `MUTATE REQUEST:`. Each request receives a new request identifier and isolated graph thread, clearing prior retrieval evidence so that an earlier tool trace cannot authorise a later write. The Memory model has four tools: retrieve,

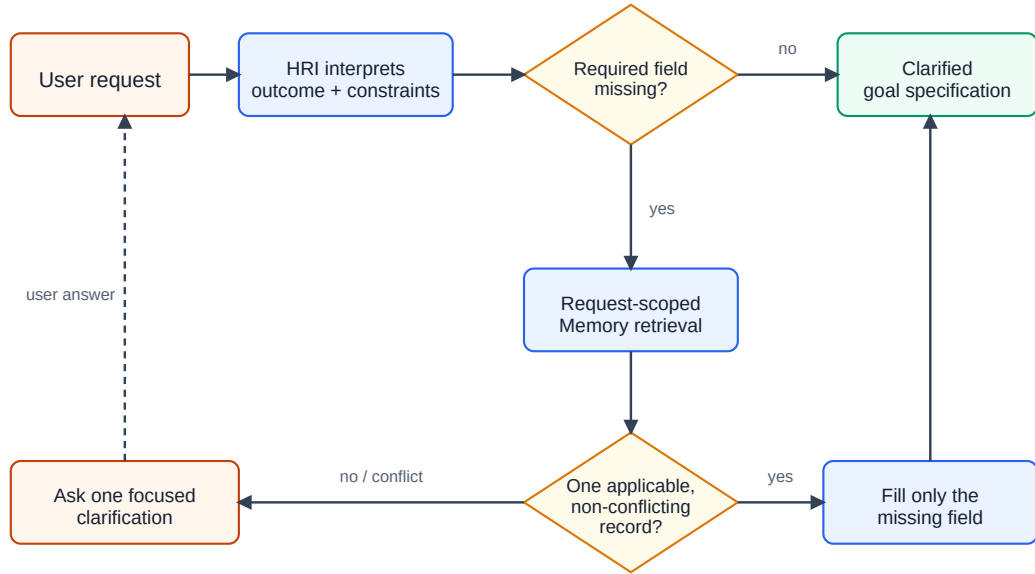


Figure 5: HRI intent-resolution subgraph. Memory may supply only a missing field when one applicable non-conflicting record exists; otherwise HRI asks one focused clarification.

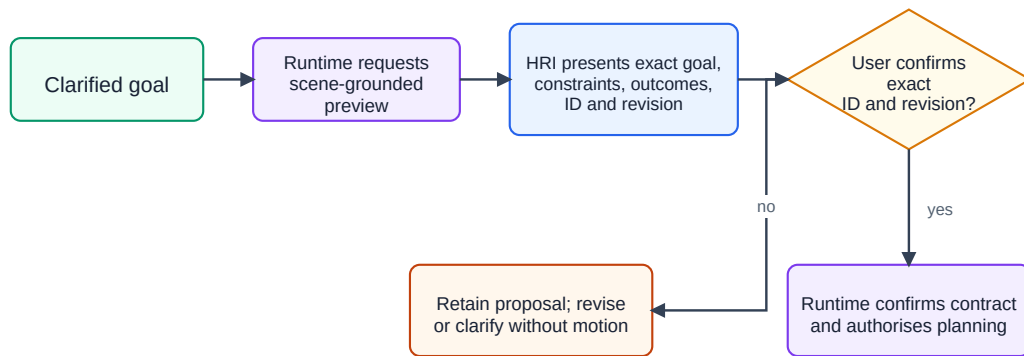


Figure 6: HRI preview-and-confirmation subgraph. The HRI model presents the contract, while Runtime alone validates the exact goal identity and permits planning.

remember, update, and forget. Its graph alternates between one model node and a tool node until the model returns a final JSON response.

Two-stage retrieval. As illustrated in Figure 7, the model first generates focused query phrasing. EmbeddingGemma encodes each query; cosine similarity is computed against the normalised store. Each query contributes its configured top- k candidates. Host code then applies deterministic ranking, deduplication by memory identifier, maximum-similarity merging, and a global cap. The current measured configuration uses $k = 3$ and $\text{cap} = 3$. The semantic model receives only this bounded set and may return only unchanged records whose identifiers, text, and similarity match the candidates. This separates high-recall candidate construction from task-specific applicability judgement.

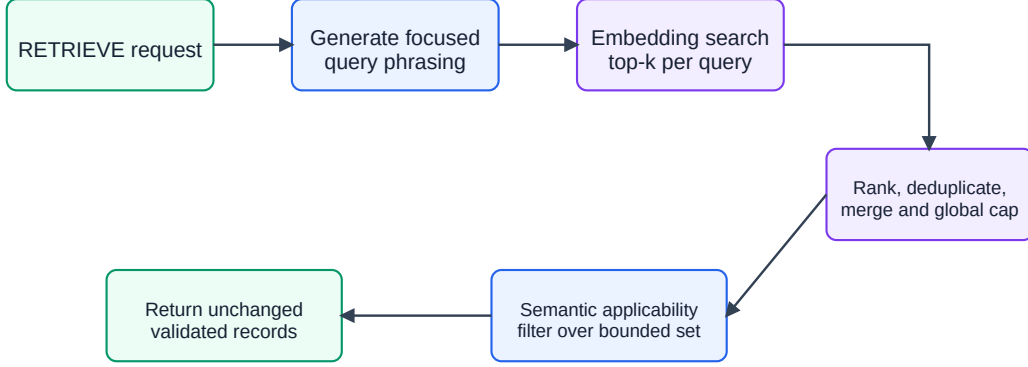


Figure 7: Memory retrieval subgraph. Deterministic candidate construction bounds the records passed to the semantic applicability filter.

Mutation policy. A write tool is invalid unless the current request is labelled as mutation and retrieval has already occurred. Update and forget targets must be among the records retrieved for that same request. The model may perform at most one justified remember, update, or forget operation; otherwise it returns a no-op or error. HRI separately enforces the future-facing consent policy: exact confirmation of the current physical goal is not memory consent. Text records and their regenerated embeddings are persisted together as the preference store, and optional trace sinks record queries, scores, caps, returned IDs, and timing without changing the response.

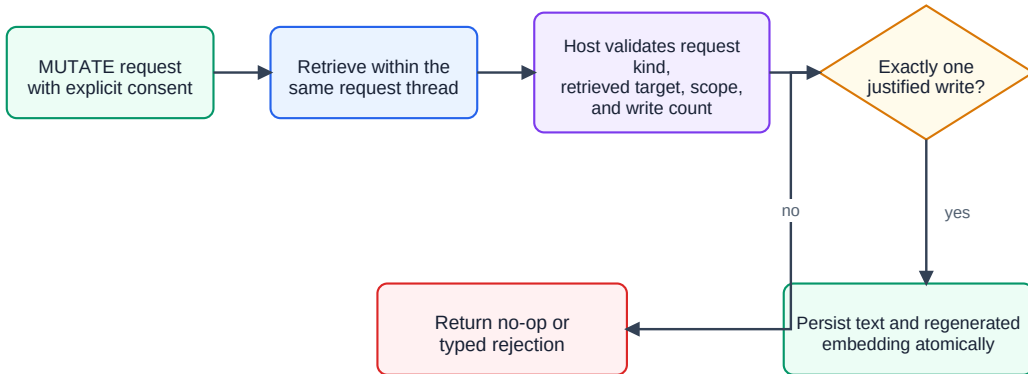


Figure 8: Memory mutation subgraph. Retrieval must occur within the same isolated request before host code permits at most one justified write.

The design nevertheless relies on free-text records and model-mediated mutation identity. The longitudinal experiment tests creation, immediate use, update, delayed retrieval, scope isolation, and revo-

cation and exposes cases in which semantically related writes can target the wrong identity. This is a measured limitation and motivates future structured user, task, and preference- dimension keys.

3.9 Planner Agent Design

Planner is a stateless, scene-grounded VLM boundary with two typed modes. It has no tools, persistent conversation, publication authority, or mutable plan queue. Every call contains a system prompt, one current image, and one explicit request payload. Temperature is zero, and returned text must parse into the requested contract; invalid shapes receive only a bounded targeted schema- correction attempt before Runtime treats the call as an error.

Preview mode. Before confirmation, Planner receives the clarified goal, explicit constraints, optional guidance, and a fresh frame. It returns a `GoalProposal` whose status is `READY`, `ALREADY_SATISFIED`, or `BLOCKED`. A ready proposal includes the exact high-level outcome, constraints, observable final results, and a short nominal outline. The outline communicates strategy to the user but is neither frozen nor executable. Runtime, not Planner, assigns the goal identifier and revision.

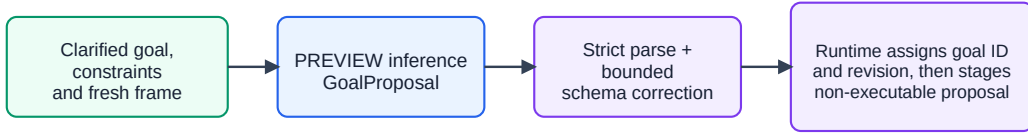


Figure 9: Planner preview subgraph. The returned outline is non-executable; Runtime parses it and assigns the trusted goal identity.

Execution-cycle mode. After confirmation, Planner receives the immutable `GoalContract`, a strictly fresh frame, the cycle trigger, ordered terminal history, and optional operator guidance. It returns `ACT`, `REQUEST_FINAL_VALIDATION`, `BLOCKED`, or `NEEDS_USER_INPUT`. An action may contain up to three candidate tasks, but the controller publishes only the first and discards the prediction tail. The next call therefore reasons from the scene actually produced by execution rather than continuing a stale predicted queue. Figure 10 shows this execution-cycle boundary separately from preview.

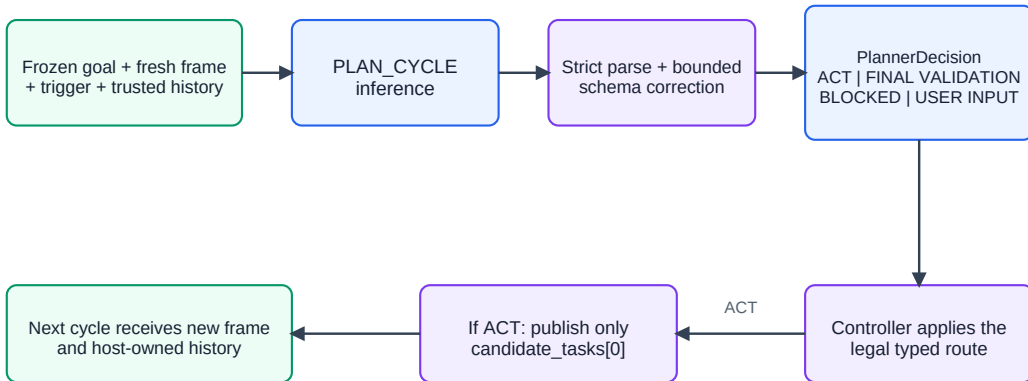


Figure 10: Planner execution-cycle subgraph. Planner proposes a typed route; Controller applies it and, for `ACT`, publishes only the first candidate before collecting a fresh frame and trusted history.

Planner cannot weaken the goal after failure, mark execution complete, or infer success from the number of attempted actions. It sees Monitor and Validator outputs only after Controller has converted them into trusted history. Its fresh-frame, history-use, grounding, capability-fence, and completion-routing behaviour are evaluated by the Planner component studies, named interfaces, workflow ladder, and replanning ablations. Deterministic replanning ablations justify the control policy but are not evidence of learned Planner quality.

3.10 Monitor Agent Design

Monitor is an asynchronous observation service for exactly one active **STEP** publication. Publishing a task increments a service generation, replaces the single active slot, and starts the worker if necessary. The worker captures the latest **prefmem** frame at a bounded interval and begins no inference unless its timestamp and frame sequence are newer than the publication. Replacing or retiring the task invalidates the generation, so a late callback cannot mutate the replacement publication.

The model prompt contains the publication, frozen goal identity and revision, step ID, instruction, ordered expected-observation criteria, and known failure conditions. It returns criterion states, a grounded observation, overall **ONGOING**, **SUCCESS**, or **FAIL**, an optional failure report, and an emergency envelope. Host parsing requires the exact ordered criterion IDs. A camera, model, parser, or callback error is reported as a service error and retires the observation job; it is never relabelled as a physical task failure.

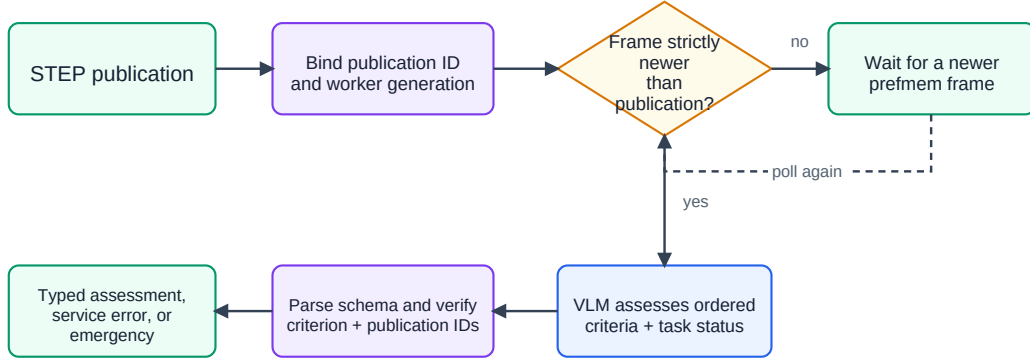


Figure 11: Monitor observation-service subgraph. The learned assessment runs only on a frame that is strictly newer than the bound publication.

A valid emergency envelope bypasses ordinary task-result parsing so visible danger cannot be hidden by an in-flight replacement. Otherwise Runtime accepts only an assessment that still belongs to the active publication. Controller then applies deterministic temporal aggregation. By default, success requires two confirmations spanning at least two seconds and failure requires two confirmations. An increase in the number of met criteria refreshes the no-progress clock; an inconclusive view does not itself erase prior candidate success unless a visible contradiction appears. In MuJoCo mode, terminal Monitor evidence is deferred until the executor reports that motion has settled.

Monitor is deliberately not the whole-goal authority. It determines whether the current instruction appears ongoing, successful, or failed. The component and interface studies test classification, temporal stability, progress, errors, stale events, and Planner hand-off, while EXP-06 compares production Monitor with a no-visual-advancement control. These results must separate learned visual recognition from deterministic host timing and fencing.

3.11 Validator Agent Design

Validator implements two workflows around one immutable **ValidationContract**. During exact confirmation, it receives the **GoalContract** and confirmation frame and expands every broad final outcome into ordered detailed visible criteria. Contract construction validates that each detailed criterion maps

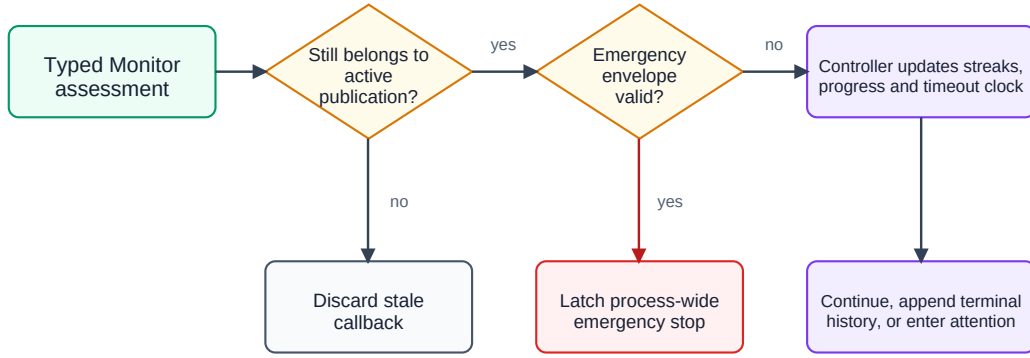


Figure 12: Monitor host-aggregation subgraph. Runtime rejects stale callbacks or latches valid emergency evidence before Controller updates progress and terminal-stability state.

to exactly one broad outcome and that every broad outcome is covered. Compilation must finish before planning is authorised. The resulting checklist is frozen with the goal so later failure cannot weaken the definition of success.

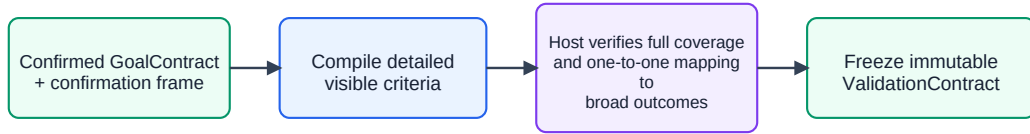


Figure 13: Validator checklist-compilation subgraph. Host validation requires complete coverage before freezing the immutable validation contract.

During execution, Planner may request final validation. Controller then creates a **FINAL_VALIDATION** publication instructing the operator to leave objects unchanged and move only the camera when needed. The Validator service accepts only a publication whose goal ID and revision match the frozen contract. Like Monitor, it polls strictly post-publication **prefmem** frames, fences service generations, and treats model/schema errors separately from physical incompleteness.

Each assessment must cover the complete ordered detailed checklist. The service accumulates evidence across views: a later **MET** or **NOT_MET** can replace an earlier **UNKNOWN**, while a new unknown view does not erase previously visible evidence. Runtime converts the accumulated assessment into a **ValidationReport**. Host aggregation maps detailed criteria back to the frozen broad outcomes and deterministically derives **COMPLETE**, **INCOMPLETE**, or **NEEDS EVIDENCE**. Unknown evidence requests camera movement only; it does not prove a task failure. Stable incomplete evidence becomes a **FINAL_VALIDATION_FAIL** history record and returns to planning. Stable complete evidence is the only ordinary path to **COMPLETE**.

Validator is independent in responsibility and context, not in model weights: it currently shares the primary Gemma endpoint with the other semantic roles. The separate frozen checklist and host aggregation prevent Planner from certifying its own task count, but correlated visual errors remain possible. Validator component studies test checklist coverage, final-state judgement, history use, missing evidence, and reset isolation; EXP-07 tests false-completion prevention, and EXP-09 exposes the cost of false rejection in a dynamic low-contrast case.

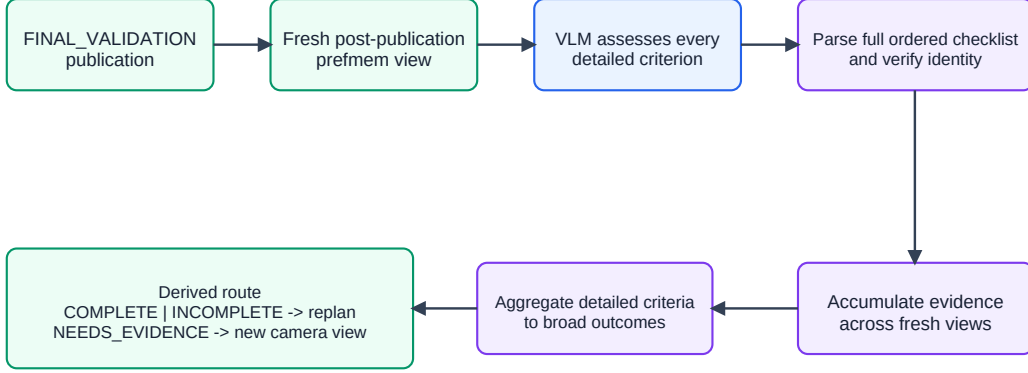


Figure 14: Validator final-assessment subgraph. Host code verifies the full ordered checklist, accumulates evidence across fresh views, and derives the final route from the frozen broad outcomes.

3.12 Execution Adapter Design

The executor is deliberately narrower than the task-level agents. It accepts only a current **STEP** publication and exposes no goal-revision, preference, confirmation, or completion authority. In human mode the adapter is the task display and a person performs the instruction. In MuJoCo mode, **ExecutionService** owns one asynchronous publication-fenced job and passes it through compilation, grounding, control, settling, and outcome recording. Figure 15 shows the strict execution path; Figure 16 isolates the experiment-only assistance path so it cannot be mistaken for ordinary execution.

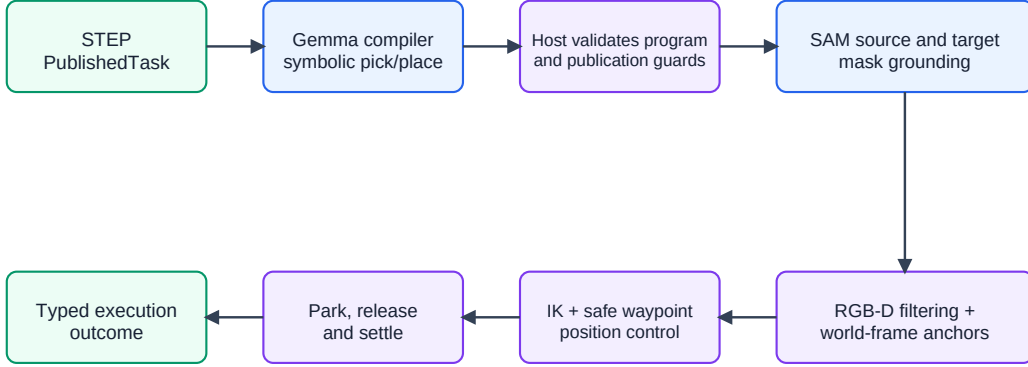


Figure 15: Strict MuJoCo execution-adapter subgraph from publication through compilation, grounding, deterministic control, settling, and typed outcome.

Gemma 4 compiles the natural-language instruction into a strict supported pick/place program. Host validation rejects unsupported operations, malformed selectors, and invalid anchors before motion. SAM 3.1 receives the robot-hidden **sam** image and returns source and target masks. Synchronized depth, calibrated intrinsics, and camera extrinsics convert filtered mask points into world-frame anchors. A damped-least-squares IK and position controller then executes a fixed safe waypoint sequence for approach, grasp, lift, transfer, place, release, retreat, and parking. Runtime treats executor activity as a liveness heartbeat but not as visual proof that the task criterion is met. After motion, the robot is parked and the manipulated object is allowed to settle before Monitor can establish a terminal result.

Cancellation and scene-change requests route the controller to a safe hold before replanning. Experiments may explicitly enable simulator-oracle grounding after a typed post-call grounding failure. The strict SAM attempt and the assisted downstream continuation remain separately recorded; an empty,

wrong, or service-invalid grounding is not silently corrected. Calibration, **C-EXE**, EXP-08, and EXP-09 decompose compiler, mask, anchor, motion, and end-to-end behaviour. The adapter is supporting infrastructure and must not be described as a learned VLA contribution.

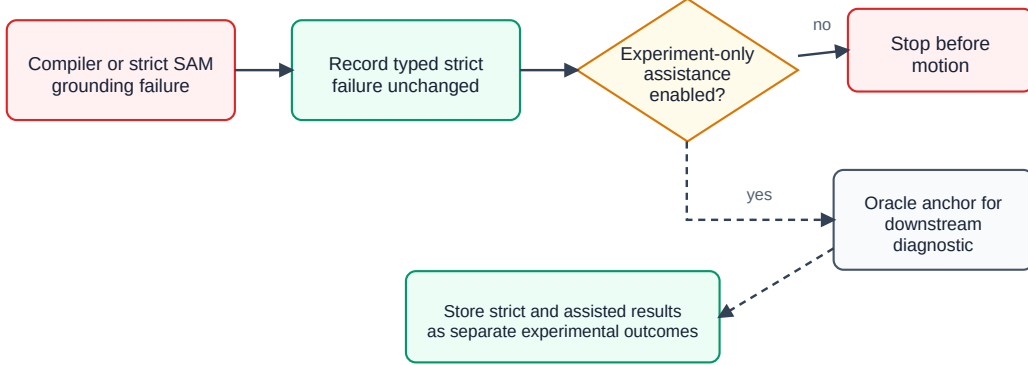


Figure 16: Experiment-only grounding-assistance subgraph. The strict failure remains recorded even when an oracle anchor permits a separate downstream diagnostic continuation.

3.13 Inter-Agent Interaction Protocol

The agents do not participate in an unrestricted shared conversation. Figure 17 therefore shows only the principal authority paths around the deterministic mediation hub; the chronological phases are expanded separately in Figure 18. HRI invokes Memory and safe Runtime tools; Runtime invokes Planner, Monitor, Validator, and Executor; Controller converts their typed results into state transitions; and Runtime returns only authoritative notifications to HRI. Planner never receives a raw Monitor or Validator callback. It receives an immutable host-owned history record on the next cycle. Monitor and Validator do not call Planner or Executor. This mediated protocol prevents a model from treating another model’s unvalidated text as authoritative state.

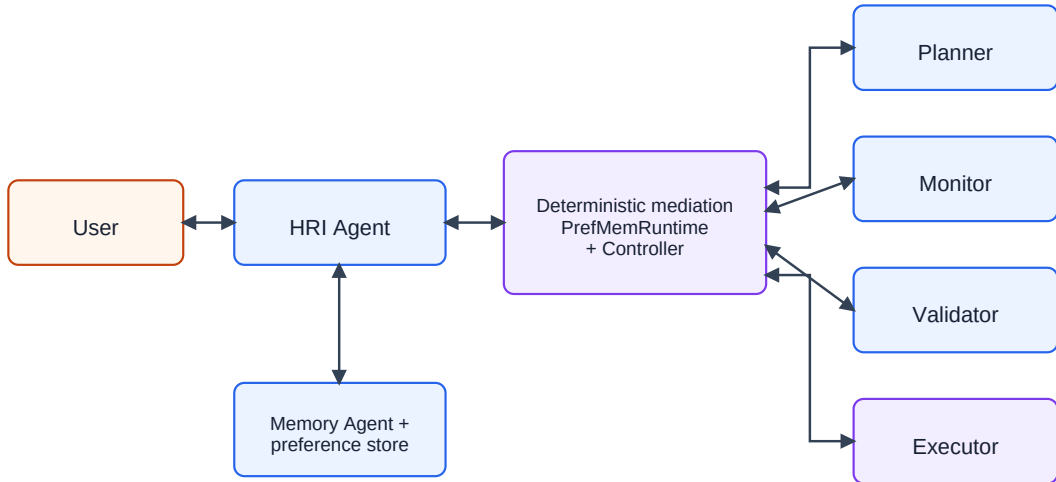


Figure 17: High-level inter-agent authority graph. Every role-specific path is mediated by Runtime or Controller; no unrestricted mutable state is shared between agents.

Appendix Table E.3 lists the corresponding typed artifacts and authority rules.

The complete interaction sequence has three phases. First, HRI and optional Memory evidence produce a preview, the user confirms its exact identity, and Validator freezes the checklist. Second,

Controller repeatedly requests a fresh Planner decision, publishes one task, and records stable Monitor evidence. Third, Planner requests final validation, Validator accumulates checklist evidence, and host aggregation either completes the goal, returns an incomplete record to planning, or asks for a new camera view. Attention and emergency paths can interrupt any active publication without altering the confirmed goal.

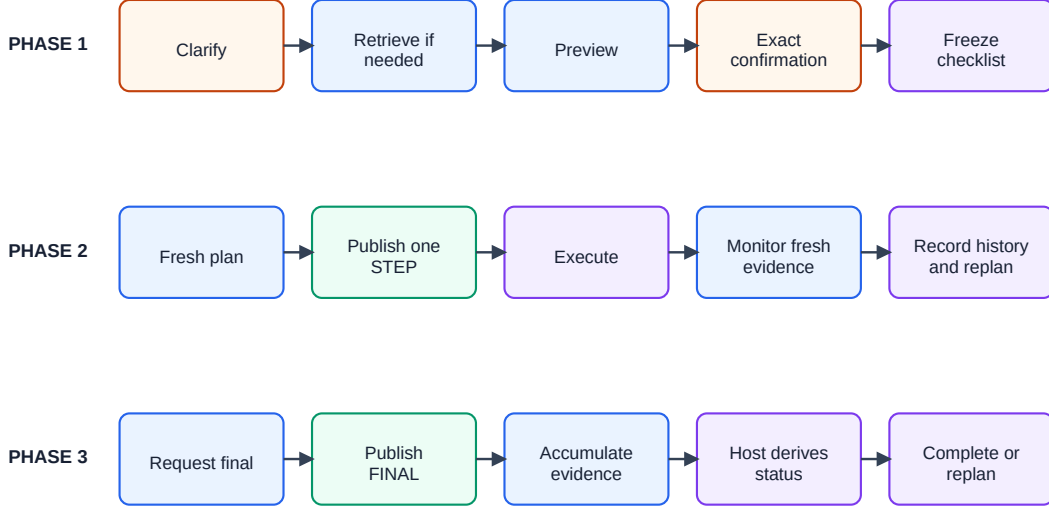


Figure 18: Expanded three-phase interaction sequence: contract formation, receding-horizon execution, and independent final validation. Attention and emergency routes may interrupt either active phase.

3.14 Deployment and Resource Model

The five logical agents share one primary reasoning endpoint rather than loading five model copies. Development model selection favoured a mixture-of-experts model near 30 billion total parameters and approximately 4 billion active parameters as a practical balance between semantic capability and inference speed. PrefMem therefore uses the instruction-tuned Gemma 4 26B A4B model, whose model card reports approximately 27 billion resident parameters and a 4-billion-active MoE architecture (Gemma Team 2026). Active parameters reduce per-token compute, but the full BF16 weights remain resident; their raw parameter storage is approximately 54 GB before KV cache, vision, runtime, and allocation overhead.

The primary vLLM endpoint is configured with BF16 precision, batch size one, a 32,768-token maximum sequence, and a GPU-memory-utilisation target of 0.95. On the 96 GB RTX PRO 6000 Blackwell workstation GPU (NVIDIA 2026c), development runs used approximately 90% of device memory. The utilisation target is an allocator configuration rather than a scientific measurement; the final reproducibility package should retain the exact launch command and profiler or `nvidia-smi` evidence. KV-cache memory grows with sequence length and is a major serving constraint (Kwon et al. 2023), reinforcing the decision to provide each logical role with targeted context rather than a union of all histories and tools.

The longer-term design aim is local execution on one edge compute node, but the current development topology retains separately configurable EmbeddingGemma and SAM endpoints. Consolidating every service on one GPU is therefore a target, not a demonstrated property of the evaluated system. The RTX PRO 6000 is a workstation-class GPU rather than a consumer device. Jetson AGX Orin offers at most 64 GB in its documented configurations (NVIDIA 2026a), while Jetson Thor modules offer larger but SKU-dependent edge-memory budgets (NVIDIA 2026b). Full-BF16 Gemma deployment on smaller embedded targets would require quantisation, offloading, shorter context, or a smaller model.

A future learned VLA will add action-model weights, visual tokens, action cache, and real-time control constraints. The available context or precision budget may consequently need to shrink. The narrow `PublishedTask` interface and role-specific contexts allow this executor to be integrated without

expanding the HRI, Memory, Monitor, or Validator prompts. The internal model- selection sweep was an engineering activity rather than a registered experiment; the thesis should not generalise its choice as a universal optimum without a matched accuracy, latency, token, and peak-memory comparison.

3.15 Recovery and Software-Safety Boundary

Model, camera, schema, publication, and service errors enter an explicit attention state with their reason preserved. When a current publication exists, the operator may resume it with a new attempt identity or request replanning with guidance. Resumption republishes the same instruction under a new attempt identity; operator-requested replanning retires the current publication and records an interruption when applicable. Neither route changes the frozen goal.

The default `AUTO_REPLAN` policy applies only to ordinary step publications. Runtime defers the visual no-progress timeout while the lower- level executor is still preparing or moving. At timeout, Controller retires the exact publication, records an `INTERRUPTED` outcome with typed timeout context and last accepted evidence, captures a strictly newer frame, and starts a new planning cycle. Consecutive and per-normalised-instruction timeout guards bound automatic recovery. Final-validation timeouts, an attention-gate policy, exhausted guards, or failure while retiring and replanning enter attention; timeout is never fabricated as Monitor `FAIL` because lack of progress is not visual proof of physical failure.

Open-category goals may use a frozen `DynamicObjectScope` whose membership remains live. A confirmed new object during motion requests cancellation and safe hold before replanning. Stack collapse or regression is handled through ordinary Monitor and Validator evidence rather than privileged simulator state.

Monitor, Validator, or the operator may also request the shared emergency latch. This stops further publication in the current process and invokes the configured stop hook once. The mechanism is a software control boundary, not a safety-rated emergency system; the current hook does not constitute real-robot safety certification.

3.16 Requirement-to-Evidence Traceability

Table E.4 maps the design requirements to their principal implementation and evaluation families. It prevents a learned component from receiving credit for deterministic host behaviour and prevents a successful physical state from hiding an upstream semantic error. Exact results and uncertainty are deferred to the Results chapter.

The complete mapping is provided in Appendix Table E.4.

4 Experiments and Evaluation

4.1 Evaluation Objectives and Claim Boundary

The evaluation asks whether PrefMem’s explicit preference, confirmation, replanning, monitoring, and validation boundaries produce the behaviours for which they were designed. It does not attempt to estimate a single notion of “robot intelligence.” The five research questions from Section 1.2 are operationalised as distinct outcomes: autonomous resolution of an omitted field, abstention under missing or conflicting evidence, explicit-current-over-historical precedence, preference lifecycle integrity, correct task-level routing, and confirmed physical goal achievement. Appendix Table A.2 maps each question to its primary experiment, comparison, inferential unit, and outcome.

The complete catalogue contains 107 finished experiments: 14 readiness and calibration studies, 32 isolated component studies, 12 named-interface studies, seven cumulative workflow studies, nine ablations, 24 original end-to-end cases, and nine focused thesis experiments. Here, *finished* means that the registered observations and report exist; it is not a performance label. The run accounting is provided in Appendix Table A.1. Because these layers have different units, controls, and purposes, they are not pooled into a suite-wide pass rate.

The focused experiments, EXP-01 through EXP-09, provide the principal evidence for the thesis claims. Readiness, component, interface, workflow-ladder, and original end-to-end results instead establish measurement validity, expose integration mechanisms, or motivate the focused contrasts. The evaluation is development and non-confirmatory unless a particular focused protocol and freeze file explicitly fixed its cases, conditions, and scoring before execution. No experiment compares PrefMem with a trained VLA or a monolithic all-tools agent. The evidence can therefore support boundary-specific

claims about PrefMem in the registered tabletop domain, but not VLA superiority, general household competence, or universal benefit from multi-agent decomposition.

4.2 Experimental Platform and Fixed Configuration

Experiments use the fixed TB6C-v2 MuJoCo scene: a Franka Emika Panda, one white board, one cyan board, and red, green, blue, yellow, purple, and orange cubes. Semantic agents receive the registered third-person `prefmem` view, which includes the robot. The execution adapter receives the synchronised robot-hidden `sam` RGB-D view. Hidden simulator state is unavailable to the agents and is used only to verify resets, score placement geometry, and adjudicate physical outcomes. This creates a separation between ordinary visual evidence and scoring authority.

The evaluated semantic roles share the primary Gemma 4 reasoning endpoint but retain the role-specific prompts, schemas, images, and tool surfaces described in Section 3. EmbeddingGemma supplies normalised retrieval vectors; SAM 3.1 supplies source and target masks; calibrated RGB-D produces world-frame anchors; and the deterministic Panda controller executes supported pick/place programs. Model version, decoding configuration, prompt and schema hashes, camera identity, scene/reset identity, and controller configuration are recorded with each applicable attempt.

Unless an ablation changes a named policy, the full workflow uses two consistent assessments for stable success or failure, a 30-second ordinary STEP no-progress timeout, a 20-cycle goal guard, and a two-consecutive-timeout automatic-replan guard. Executor-active intervals are exempt from the visual no-progress timeout. Paired conditions share the same language context, scene state, frame sequence, timing, and host configuration; only the registered treatment boundary differs.

4.3 Evaluation Design

4.3.1 Evidence Layers

The suite follows a layered evidence strategy. Readiness and calibration first test the scene, camera, reset, logging, deterministic safety routes, selector coverage, anchor geometry, and controller/release bias. Isolated component experiments replace unrelated learned dependencies with deterministic fixtures, so that one model-facing contract can be examined without another learned role hiding its errors. Named-interface studies restore exactly one interaction; the cumulative ladder then records integration behaviour as requirements are added. Paired ablations estimate selected boundary effects. Finally, focused experiments exercise preference decisions, longitudinal state, feedback gates, registered perturbations, and the complete simulated path. Detailed results for the supporting layers are in Appendices B and C.

This hierarchy prevents two attribution errors. First, a deterministic timeout or stale-event guard is not credited to a learned Monitor or Planner. Second, a correct physical placement cannot hide an incorrect goal contract, wrong target query, assisted grounding call, or false completion decision. Each layer is scored only at the responsibility boundary it validly exercises.

4.3.2 Trial Units and Conditions

One immutable trial identity contains the case and aim, condition, scene/reset hash, exact utterance, memory snapshot, model and decoding configuration, prompt and schema hashes, random seed, source version, and artifact profile. Changing one member produces a new trial rather than silently modifying an old result.

Original component and interface aims normally use eight semantically distinct contexts with five fixed variants, producing 40 observations. The variants change phrasing, object order, irrelevant clauses, seed, or registered camera perturbation; they are not treated as five independent environments or users. Original end-to-end cases use five variants per case. EXP-01 through EXP-07 use paired clustered designs: three surface variants are repeated observations, while the context, scenario, or lifecycle is the inferential unit. EXP-08 uses three fresh language/seed pairs in each of six families, and EXP-09 uses three pairs in each of four contract types. These small final studies are reported descriptively rather than as population reliability estimates.

4.3.3 Invalid Runs, Retries, and Provenance

Every attempt is retained. A valid but unfavourable response is a capability observation and is never rerun to obtain a preferred answer. An unplanned infrastructure interruption or invalid fixture may be

followed only by an exact-tuple attempt after repair; the original attempt, invalidity reason, repair, and retry lineage remain in the audit record. Focused experiments prohibit capability retries and response reuse. A deliberately injected outage is a valid stimulus, not an infrastructure-invalid run.

The experiment status is therefore separate from behavioural quality. Reports preserve direct predicates, formal historical envelopes where present, model calls, tokens, timing, physical-action counts, and evidence paths. Machine- readable observations and human-readable reports remain in the versioned experiment-result directory; representative structured outputs are reproduced in Appendix D.

4.3.4 Strict and Assisted Physical Evidence

Strict grounding requires every source and target selector in an episode to be resolved through SAM and RGB-D. When a frozen physical experiment explicitly permits simulator-oracle continuation after a typed grounding failure, the strict failure remains recorded and the downstream assisted result is reported separately. Assistance exposes motion, monitoring, or validation behaviour; it does not convert the strict observation into a pass. Simulator truth likewise scores physical state but is never supplied to the semantic agents as task evidence.

4.4 Metrics and Statistical Analysis

Primary outcomes are direct binary or count predicates rather than a global experiment flag. Preference studies distinguish autonomous correct resolution from correctness after an allowed clarification budget. Conflict studies score pre-resolution commitment, focused clarification, and post-answer correctness. Execution-loop studies score the registered host route, endpoint advancement, discrepancy recovery, false completion, false rejection, and correction request. Physical studies separately score first-action semantics, action count, strict grounding, simulator-truth goal state, safety, Validator finalisation, and the complete registered chain.

For a paired clustered binary outcome, let $Y_{cvk} \in \{0, 1\}$ denote the observation from cluster c , surface variant v , and condition k . The condition mean within a cluster and the paired risk difference are

$$\bar{Y}_{ck} = \frac{1}{V_c} \sum_{v=1}^{V_c} Y_{cvk}, \quad \hat{\Delta} = \frac{1}{C} \sum_{c=1}^C (\bar{Y}_{cT} - \bar{Y}_{cC}), \quad (2)$$

where T and C denote treatment and control. Cluster-bootstrap intervals resample clusters rather than individual paraphrases. Exact cluster sign-flip tests are secondary when registered and their assumptions hold. Original matched ablations retain paired risk differences and exact McNemar tests; unpaired descriptive rates retain direct counts and Wilson intervals where available. Latency, model calls, prompt burden, and physical actions are secondary resource outcomes.

Placement scoring uses the settled hidden object centre. For registered target (x^*, y^*) , radial error is

$$e_r = \sqrt{(x_{\text{settled}} - x^*)^2 + (y_{\text{settled}} - y^*)^2}. \quad (3)$$

Signed axis errors are retained because a small radial value can hide a repeatable directional bias. EXP-08 and EXP-09 intervals are descriptive due to their three observations per family or contract type. No multiplicity- adjusted suite-wide hypothesis is constructed, and no denominator combines readiness rows, dialogue observations, retrieval-grid rows, and physical episodes.

4.5 Readiness and Supporting Diagnostic Results

4.5.1 Scene, Camera, Reset, and Safety Readiness

The readiness layer established that the frozen fixture and evidence pipeline were usable for development experiments. Scene resets reproduced the registered state; all objects and analytical waypoints were visible or reachable in the specified distribution; camera projection/deprojection error was negligible; the reset verifier rejected a displaced-object negative control; and the eight replanning plus twelve safety replays followed their registered deterministic routes. These results validate measurement and host ordering, not learned semantic competence or real-robot safety. Complete counts and qualifications are reported in Appendix Table B.1.

4.5.2 Grounding and Placement Calibration

Calibration separated three mechanisms. Projection and oracle-mask anchor estimation were sub-millimetre at their principal quantiles. SAM masks were high quality conditional on detection, but detection covered only 355 of 520 selector-position observations. Oracle-coordinate and full-path placement both retained an approximately +4 mm signed x -shift, localising much of the bias to grasp, controller, release, or contact dynamics rather than camera geometry. The stage-specific results are in Appendix Table B.1; consequently, no global perception correction is claimed.

4.5.3 Component and Interface Diagnostics

The broad component studies show why later focused endpoints were required. HRI was strongest on typed runtime state and preference policy but weaker on unresolved material ambiguity. Planner used trusted history and completion routes reliably but showed lower spatial-grounding and capability-fence accuracy. Deterministic Memory candidate construction was correct in 40/40 rows, whereas four original semantic Memory aims stopped before inference due to a checkpoint/thread configuration defect. Monitor and Validator preserved several host invariants but remained visually overconfident or insufficiently uncertain in difficult states. Compiler permissiveness, SAM coverage, and IK exceptions formed separate executor limitations.

The cumulative ladder is retained as an integration diagnostic, not a causal component ranking, because each level adds both a component and additional required predicates. The original Memory level in particular exposed a harmful integration seam rather than proving that preference memory is intrinsically harmful. Appendix Table B.2 reports the direct component results and their interpretation boundaries.

4.6 Preference-Aware Intent Evaluation

4.6.1 Applicable Scoped Memory

EXP-01 compares the same under-specified utterance with applicable scoped Memory, empty Memory, and an exact-instruction ceiling. Scoped Memory produced autonomous correct resolution in 36/36 observations versus 0/36 with empty Memory, a paired difference of +1.00 with all twelve cluster differences positive. Both conditions nevertheless reached 36/36 correct goals within the two-turn budget. The contribution is therefore a one-turn reduction in interaction cost, not an increase in attainable correctness. Only 9/36 populated observations returned a perfectly scoped candidate set, so the effect belongs to the combined HRI-Memory decision boundary rather than raw retrieval precision. Condition tables and plots are in Appendix Table C.1 and Figure C.1.

4.6.2 Conflict-Aware Abstention

EXP-02 tests whether caution is selective rather than universal. Two equally scoped conflicting records caused no pre-resolution goal in 36/36 observations, focused clarification in 36/36, and the correct post-answer goal in 36/36. A unique applicable record remained autonomous in 35/36. The system therefore distinguished unique evidence from conflict instead of simply selecting the top retrieved item. However, wrong-task candidates were excluded in only 21/36 unique-Memory observations; downstream reasoning compensated for noisy retrieval rather than eliminating it. Detailed condition and resource outputs are in Appendix Table C.1.

4.6.3 Explicit Current-Instruction Override

EXP-03 and EXP-04 test field-level composition: Memory may supply an omitted usual cube, but the board explicitly named in the current instruction must not be replaced by a historical board. The original EXP-03 measurement was sensitive to how yes/no clarification was observed and is preserved unchanged. The held-out EXP-04 replication used new users, tasks, wording, and cube assignments with a corrected clarification observer. In the opposing-Memory condition, the explicit board was preserved in 35/36, the correct goal was reached within budget in 35/36, and the historical board was selected in 0/36. The result supports current-over-historical precedence at the tested fields, while the remaining wrong-task retrievals constrain any claim about retrieval scope. Appendix Table C.1 retains both the original and replication results.

4.6.4 Longitudinal Preference Lifecycle

EXP-05 carries eight independent stores through 30 ordered interactions without an in-lifecycle reset. Initial creation, immediate direct recall, and initial HRI use were 8/8. Explicit update and updated-value recall were 7/8, while updated HRI use was 5/8. After fourteen semantically related distractor mutations, delayed direct recall was 2/8, delayed HRI use was 3/8, and final direct recall remained 2/8. Store serialization stayed consistent in 240/240 interactions and the old value remained absent after update, but free-text mutation arbitration sometimes reused or replaced the wrong semantic identity.

This is a decisive negative result for unrestricted long-term-memory claims. The prototype supports a clean short lifecycle but does not robustly preserve a focal preference under repeated related mutations. The full trajectory is in Appendix Table C.2 and Figure C.2; it motivates immutable structured user, task, preference-dimension, and provenance keys.

4.7 Execution-Loop Evaluation

4.7.1 Fresh Replanning

The stable-failure ablation recovered 40/40 deterministic traces with fresh replanning and 0/40 with a frozen action queue. The timeout-policy ablation recovered 40/40 under both automatic and attention-gated policies, but automatic replanning required zero operator interventions rather than 45 and introduced no registered stale action or guard breach. These experiments justify retiring stale task queues and show an interaction-cost benefit at the deterministic host-policy boundary. Because the Planner outputs were fixtures, they do not establish learned Planner quality. Appendix Table B.3 contains the paired results.

4.7.2 Monitor Ablation

EXP-06 holds the host timing, publications, frames, and scenario clusters fixed while replacing production Monitor with a control that never claims visual advancement. Production Monitor produced the expected host route in 22/36 observations versus 12/36 for the control, a +27.8 percentage-point paired change with a cluster-bootstrap 95% interval from +8.3 to +50.0 points. It advanced 7/9 stable endpoints and registered 3/3 visible-progress cases, but recovered only 3/15 visual discrepancies; seven of 78 invocation attempts were schema-invalid. Host holds and timeouts were equal in both conditions and are not credited to the learned Monitor. Appendix Table C.3 and Figure C.3 separate these functions.

4.7.3 Validator Ablation

EXP-07 compares production Validator with stopping after actions and assuming success. On 21 visibly incomplete observations, production Validator produced zero false completions and requested correction in all 21; the control falsely completed all 21. On genuinely complete states, production finalised 14/15 versus 15/15 for the control. Independent validation therefore provides a large false-completion benefit with a measurable false-rejection cost. The result justifies a separate completion gate but not unquestioned Validator authority; its detailed routes and criterion accuracy appear in Appendix Table C.3 and Figure C.3.

4.8 Robustness and End-to-End Evaluation

4.8.1 Focused Competency Envelope

EXP-08 uses three fresh language/seed pairs in each of six families: slip, timeout, occlusion, planned outage, scoped preference, and explicit override. The complete registered family-specific event occurred in 15/18 episodes and the registered safety condition in 18/18. Occlusion, outage containment, and scoped preference occurred in 3/3 each; slip and timeout recovery occurred in 2/3 each. Fourteen episodes completed strictly, two used separately recorded oracle assistance, and two stopped safely at selector/grounding. The two unreached recovery stimuli failed on unsupported aliases before the intended perturbation, localising the mechanism to language-to-selector normalisation. Family-level outcomes are reported in Appendix Table C.4. With only three episodes per family, this is a demonstrated envelope rather than a broad robustness estimate.

4.8.2 Simulated Intent-to-Outcome Evaluation

EXP-09 follows twelve fresh episodes across omitted-board, omitted-cube, explicit-override, and user-scoped contracts. Expected Memory records, correct goal contracts, confirmation, and correct first compiled actions occurred in 12/12, with no wrong target-board query. All twelve reached the registered simulator-truth physical arrangement safely. Eleven remained strict through all SAM grounding calls, nine were finalised by Validator, and eight satisfied the exact one-action all-strict chain. Contract-specific results and the stage funnel are in Appendix Table C.5 and Figure C.4.

The three omitted-cube episodes illustrate why stage-level scoring is necessary. All placed blue at the cyan-board centre within the physical tolerance, but Monitor or Validator rejected the low-contrast state and caused repeated actions or attention. Thus intent and physical goal achievement were 12/12 even though terminal visual closure was 9/12. EXP-09 begins from frozen exact preference fixtures and therefore tests application of retrieved evidence, not production retrieval accuracy. Representative successful and false-negative outputs appear in Appendix D.

4.9 Threats to Validity

Internal validity. The original suite is development/non-confirmatory evidence, and some source state evolved after early source-lock hashes. Several original semantic Memory aims were blocked before inference by a configuration defect, and several controlled recovery events fired at an illegal host phase or were not reached. These observations are retained as integration diagnostics rather than relabelled learned failures. EXP-03’s clarification-observer sensitivity is reported beside, not replaced by, the held-out EXP-04 replication.

Construct validity. Some original rubrics were overly lexical; for example, semantically adequate scene descriptions could fail a required word. Sensitivity audits are reported without overwriting frozen scores. Hidden simulator pose is a precise physical authority but does not reproduce real sensor uncertainty. Expected host routes are responsibility-specific proxies and must not be interpreted as general task intelligence. Strict and assisted grounding are kept separate to prevent oracle continuation from inflating learned perception evidence.

External validity. Physical results use one MuJoCo tabletop scene, one embodiment, two boards, six cubes, and one fixed camera family. EXP-08 and EXP-09 contain only three fresh episodes per family or contract type. The work therefore does not establish real-robot safety, open-world language understanding, long-horizon household generalisation, or robust preference retention over hundreds of interactions.

Statistical validity. Surface variants are correlated repetitions, so focused uncertainty resamples clusters. Twelve clusters can identify large consistent changes but provide limited precision for heterogeneous effects; three-case families are descriptive. Multiple secondary predicates are mechanism analyses rather than independent confirmatory hypotheses, and no favourable stopping rule or capability rerun was used.

Comparative validity. No trained VLA or single-agent all-tools baseline was evaluated. The evidence supports the selected architecture through boundary-specific effects, containment, and failure localisation, not through an empirical claim that multi-agent graph/loop engineering always outperforms a monolithic system.

4.10 Evaluation Summary

The evaluation answers the research questions with a mixed but coherent result. Applicable preference evidence reduced dialogue by one turn without changing attainable correctness. Conflicting evidence caused abstention, and a held-out replication preserved explicit current fields over history. The longitudinal study rejected a strong robust-memory claim by exposing free-text identity loss. Fresh replanning, Monitor, and Validator produced distinct benefits at their own boundaries, while Monitor discrepancy recovery and Validator false rejection remained important weaknesses. Finally, twelve focused simulated episodes preserved intent through physical outcome, but visual finalisation reduced the exact strict-chain count. PrefMem’s supported contribution is consequently an inspectable preference-and-evidence control architecture with measurable boundary effects and localised failure mechanisms, not a universal improvement score.

5 Discussion

5.1 Interpretation of the Main Findings

PrefMem addresses a task-level problem that is not solved by predicting a robot action alone: maintaining a trustworthy agreement about what the robot should do, why a historical preference is applicable, when the robot must ask, and what evidence permits completion. The evaluation supports this framing while showing that explicit boundaries do not make every learned decision reliable. Detailed measurements are reported in Section 4 and Appendices B–D; this chapter interprets their combined implications.

The strongest positive result is selective use of preference evidence. One applicable record removed a clarification turn without changing final correctness; equal conflict caused abstention; and a held-out replication preserved a current explicit board against opposing history. PrefMem therefore implements more than generic personalisation: history may fill an omitted field only when its scope and provenance justify doing so. This complements systems that infer household preferences from examples (Wu et al. 2023; Wang et al. 2025) by emphasising authority, confirmation, conflict, and revocability before action.

Fresh replanning, Monitor, and Validator also produced distinct effects. Fresh replanning retired a failed queue; Monitor supplied endpoint and progress evidence; and Validator prevented every registered false completion in visibly incomplete scenes. These roles operate at different temporal scales and prevent an action attempt, a local success, or a Planner prediction from becoming whole-goal evidence merely because it occurs late in a sequence.

The evidence does not support a claim that every additional agent improves the complete stack. The cumulative ladder was not monotonic, the old Memory integration was harmful on its development endpoint, Monitor recovered only a minority of visual discrepancies, and Validator sometimes rejected correct states. Most importantly, delayed preference recall survived mutation pressure in only two of eight stores. The project has implemented and evaluated the preference lifecycle, but has not solved robust long-term personal memory.

The focused end-to-end study illustrates the value of staged evaluation. All twelve episodes formed the intended contract and reached the registered physical arrangement, whereas nine received Validator completion and eight satisfied the exact one-action strict chain. A global score would hide whether a failure arose from intent, grounding, action efficiency, physical outcome, or visual closure. PrefMem’s central systems contribution is that these stages are represented by named contracts, publications, evidence, and authorities.

5.2 Project Achievements

5.2.1 Preference-Aware Task Contracts

The first achievement is a conservative interaction policy. PrefMem retrieves before deciding, fills only a missing field from one applicable record, asks under conflict, preserves current explicit values, and separates physical goal confirmation from consent to change future memory. Embedding similarity is candidate generation rather than authority; HRI still evaluates user, task, field, and wording. This is an appropriate default for manipulation because a wrong physical commitment is costlier than one focused question.

The tested ambiguities concern known cube, board, and order fields rather than open-world intention understanding. Nevertheless, they represent a practical household interaction pattern: users omit habitual details. The project shows that these omissions can be governed through explicit scope, provenance, and abstention instead of an undocumented prompt convention.

5.2.2 Inspectable Graph and Loop Engineering

The second achievement is an explicit control graph. HRI, Memory, Planner, Monitor, Validator, Runtime, Controller, and the executor exchange typed goal proposals, contracts, publications, assessments, history records, and notifications rather than sharing an unrestricted conversation. Models propose semantic content; deterministic host code confirms goals, publishes actions, owns history, applies guards, and establishes completion.

Loop engineering supplies temporal discipline. One task is published, only post-publication evidence is accepted, stable outcomes retire the old identity, and the next cycle uses a fresh frame. Final validation is a separate bounded loop over a frozen checklist. This extends feedback-based embodied reasoning (Ahn et al. 2022; Huang et al. 2022; Yao et al. 2023) with explicit preference provenance, publication identity, confirmation, and terminal evidence.

One all-tools agent could reproduce much nominal behaviour. Role separation is useful because each role needs different context, temporal scope, and authority: Memory sees preference tools, Planner sees the frozen goal, Monitor sees one publication, Validator sees the whole checklist, and only HRI addresses the user. This improves failure localisation and supports shorter role-specific contexts on edge hardware. It is an implemented authority architecture, not yet an empirical proof of superiority over a matched monolithic agent.

5.2.3 Independent Evidence and Completion

The third achievement is separation of progress from completion. Monitor is step-bound, Validator is goal-bound, and Controller aggregates their evidence. Executor return is not visual success; finishing an action list is not proof of the goal; and Planner cannot certify its own plan. Validator’s prevention of all registered incomplete-state completions directly supports this boundary, while its false rejection quantifies the cost of imperfect evidence.

Monitor’s contribution is smaller but distinct: it advanced endpoints and registered progress that the no-visual control could not. Deterministic holds and timeouts were unchanged, correctly locating those behaviours in host policy rather than learned vision. The architecture therefore permits both benefit and failure cost to be assigned to the responsible layer.

5.2.4 End-to-End Feasibility and Research Deliverables

The fourth achievement is a functioning preference-to-physical-state path. In the focused MuJoCo study, the system applied the expected record, confirmed the goal, issued the correct first compiler request, and reached the physical outcome in all twelve episodes; eleven remained strict through SAM grounding. The executor is supporting infrastructure, not a new VLA. Its narrow boundary makes PrefMem complementary to semantic and hierarchical VLA research (Ni et al. 2026; Shi et al. 2025; Yang et al. 2025b; Li et al. 2026).

The project also delivers typed control software, scoped Memory, human and MuJoCo adapters, calibration, bounded recovery, evidence recording, component fixtures, focused protocols, and a 107-experiment catalogue. Strict and assisted runs remain separate, invalid attempts retain lineage, and capability failures are not rerun for replacement. Retaining positive, null, and negative evidence is itself important for an auditable systems contribution.

5.3 Shortcomings and Failure Analysis

5.3.1 Memory Identity and Scope

The most serious shortcoming is free-text memory identity. Retrieval often returned useful records alongside wrong-task candidates, and HRI compensation does not make that retrieval precise. Mutation arbitration sometimes treated a related new preference as an update across task scope or user ownership. The store stayed structurally valid while the intended semantic identity was lost. Similarity is suitable for search but not as a primary key for ownership, dimension, supersession, or revocation; increasing store or context size would not address this mechanism.

5.3.2 Semantic and Interface Brittleness

HRI sometimes previewed unresolved requests; Planner was weaker on spatial relations and capability fencing; the compiler accepted too many unsupported instructions; and healthy Monitor calls sometimes violated the schema. Typed parsing contains malformed responses but cannot correct a schema-valid semantic error. Conversely, brittle lexical evaluation can reject a semantically useful answer. The present system still relies heavily on prompting rather than a canonical task language shared across roles.

5.3.3 Visual Feedback and Correlated Error

Monitor recovered only 3/15 registered visual discrepancies. Validator rejected some correct states, including all three low-contrast blue-on-cyan finalisations in EXP-09, causing redundant action or attention despite correct physical placement. Role separation provides procedural independence, not statistical independence: the roles share one Gemma endpoint and similar views, so correlated perceptual error remains possible.

5.3.4 Execution and Integration Boundaries

The surrogate has vocabulary, detection, and control limits. Reasonable aliases may fall outside the selector vocabulary; SAM is accurate when detected but has coverage gaps; isolated actions can encounter IK exceptions; and successful placements retain an approximately +4 mm signed x -bias. Several original Memory and recovery tests also measured configuration or event-injection seams rather than the intended learned capability. Modularity improves observability but creates interfaces that must be versioned and tested; the present design makes these failures inspectable without yet making every seam robust.

5.4 Limitations

Scope and evidence. Physical evidence uses one MuJoCo tabletop scene, one Panda, two boards, six cubes, and one camera family; it does not establish real-robot transfer or safety. Most evidence is development and non-confirmatory, with clustered paraphrases and small focused families. The results locate large effects and mechanisms rather than broad reliability.

Baselines and users. There is no trained VLA, matched monolithic agent, or model-size comparison. Deterministic replanning isolates host policy, and frozen EXP-08/09 preference fixtures do not estimate production retrieval. Scripted turns also do not measure satisfaction, trust, clarification tolerance, or consent comprehension.

Model and executor validity. All semantic roles share one Gemma-family endpoint, and the workstation result does not prove that a VLA-integrated stack fits one edge GPU. The compiler, SAM, RGB-D, and deterministic IK approximate a VLA interface but not continuous learned-action uncertainty, policy latency, or embodiment-conditioned recovery.

5.5 Future Work

Future work should proceed in the following order:

1. **Structure preference identity.** Give each record immutable user, task, dimension, object, provenance, and supersession keys. Host code should classify create, update, revoke, or conflict before a model proposes text, and writes should be transactional. A held-out EXP-05 replication should measure focal survival, scope violations, update identity, and revocation over longer histories while retaining the current 2/8 baseline.
2. **Canonicalise the task/execution language.** Map aliases, objects, target regions, relations, tolerances, and visibility requirements into a versioned ontology before compilation. An allowlist grammar should reject unsupported actions before motion and be tested on held-out aliases, not tuned against EXP-08 failures.
3. **Calibrate Monitor and Validator.** Combine VLM semantics with segmentation, depth geometry, and multiple views. Treat unknown as a calibrated outcome. A held-out low-contrast set should measure false completion and false rejection together; constrained decoding and an independent model or sensor path could reduce malformed and correlated errors.
4. **Test architecture and VLA integration.** Compare the graph with a matched all-tools agent under the same model, prompt content, calls, tokens, and GPU budget; measure accuracy, authority violations, latency, VRAM, and failure localisation. Then substitute a frozen VLA through PublishedTask without granting preference, goal-revision, or completion authority.
5. **Expand physical, human, and confirmatory evaluation.** Re-run focused cases on a supervised robot with independent pose scoring, and use a separate approved study for clarification burden, trust, and consent. Report edge-resource use and freeze broader users, tasks, layouts, histories, endpoints, and scorers before inspection, while retaining stage-specific outcomes.

5.6 Concluding Perspective

PrefMem makes preference, confirmation, planning, monitoring, and validation explicit contracts around an executor. Evidence supports selective memory use, replanning, completion gating, and simulated feasibility while exposing memory and visual limits. The contribution is an inspectable authority-and-evidence architecture.

A Experimental Design and Run Accounting

Core results are summarised below. Complete experiment reports and artifacts are available in the project repository: <https://github.com/KasLoot/RoboPref>.

A.1 Run Accounting

Table A.1: Experiment-suite run accounting. A finished experiment has a complete registered observation set and report; the term is not a quality verdict.

Layer	Purpose	Count	Finished	Interpretation
Readiness and calibration	Validate fixture, evidence, geometry, and execution measurements	14	14	Measurement and mechanism evidence; not semantic performance
Isolated components	Exercise one role contract with fixture replacements	32	32	Component diagnostics with responsibility-specific predicates
Named interfaces	Restore exactly one agent/host interaction	12	12	Propagation and containment evidence
Workflow ladder	Add cumulative requirements across ambiguity contexts	7	7	Integration diagnostic; adjacent rates are not pure component effects
Ablations	Compare selected paired boundaries and host policies	9	9	Direct effects only at the frozen replacement boundary
Original end to end	Nominal, ambiguity, and recovery development cases	24	24	Broad development evidence with mixed upstream/downstream failure
Focused thesis studies	Paired claim studies, robustness envelope, and fresh simulated chain	9	9	Principal thesis evidence
Total		107	107	No suite-wide success percentage is defined

A.2 Research-Question Traceability

Table A.2: Research questions, primary evidence, and outcomes.

RQ	Operational question	Primary evidence	Inferential unit	Primary outcomes
1	Does one applicable scoped preference resolve an omitted field without dialogue?	EXP-01	12 context clusters	Autonomous correct goal, within-budget correctness, clarification turns, scope, and safety
2	Does the system abstain under conflict and preserve current explicit fields?	EXP-02, EXP-03, EXP-04	12 clusters per study	Pre-resolution commitment, clarification, explicit field, historical selection, and final goal
3	Does preference state survive create, update, delay, distraction, and revocation?	EXP-05	Eight independent lifecycles	Direct recall, HRI use, identity, scope, retention, revocation, and store consistency
4	Do fresh replanning, Monitor, and Validator improve their registered routes?	AB-RPL-F/T, EXP-06, EXP-07	40 paired traces or 12 scenario clusters	Recovery, interventions, expected route, false completion, correction, and false rejection

Table A.2: Research-question traceability (continued).

RQ	Operational question	Primary evidence	Inferential unit	Primary outcomes
5	Can a confirmed contract reach a safe simulated physical goal?	EXP-08 and EXP-09	Six families or four contract types	Registered perturbation route, strict/assisted grounding, physical state, safety, validation, and exact chain

B Supporting Readiness, Calibration, and Diagnostic Results

B.1 Readiness and Placement Calibration

Table B.1: Readiness and calibration results.

Study	Rows	Direct result	Interpretation boundary
PF-ASSET	–	Canonical, relocated, and archived scene dependencies and hashes were valid.	Reproducible loading only.
PF-SCENE	20	All eight objects were visible; all 28 object pairs remained separated; 32 analytical waypoints were reachable.	Fixture distribution, not general scene coverage.
PF-CAMERA	20 pairs	Approved calibration and aligned depth were retained; maximum round-trip error was approximately 1.14×10^{-13} px.	Camera geometry is not a plausible source of millimetre-scale bias here.
PF-RESET	20	Canonical resets reproduced; a displaced-cube negative control was rejected before restoration.	Reset-verifier evidence only.
PF-MOTION	32	All analytical approach/contact waypoints satisfied IK and workspace fences without actuation.	Does not test grasp or release.
PF-SELECTOR	160	Intended selectors were detected in 20/20 registered perturbations per selector; median radial anchor errors were below 5 mm.	Narrower and easier than the CAL-X-03 grid.
PF-REPLAN	8 traces	Every deterministic replay followed its expected fresh-request, retirement, reset, or incompleteness route.	No learned Planner calls.
PF-SAFETY	12 traces	Stale events, errors, guards, attention, and emergency latching were contained as registered.	Software invariants, not safety certification.
PF-LOG	3 attempts	Final retained attempt produced a complete artifact bundle and a 3.18 mm oracle-coordinate placement.	Artifact-pipeline demonstration, not a success rate.
CAL-X-01	375	Median signed error was +0.015 mm x , +0.055 mm y ; p95 absolute axis error was approximately 0.689 mm.	Projection is not the dominant (+ x) mechanism.
CAL-X-02	520	Oracle-mask anchors were complete; p95 absolute error was 0.137 mm (x) and 0.662 mm (y).	Accuracy conditional on the correct mask.
CAL-X-03	520	SAM detected 355/520; 165 were usable zero-detection responses. Conditional mask IoU had minimum 0.952 and median 0.997.	Main limitation is selector-position coverage, not successful-mask accuracy.
CAL-X-04	300	All oracle-coordinate placements completed; median error was +3.938 mm x , −0.012 mm y ; p95 radial error 4.180 mm.	Localises the repeatable shift to controller/release dynamics.
CAL-X-05	300	Full-path placements completed; median error was +3.996 mm x , −0.700 mm y ; p95 radial error 4.594 mm.	Sequential development run; does not erase CAL-X-03 coverage gaps.

B.2 Component and Interface Diagnostics

Table B.2: Selected component and integration diagnostics.

Boundary	Direct observations	Strength	Important limitation
HRI	State reporting 39/40; preference policy 33/40; exact tool route 32/40; registered broad ambiguity behaviour 21/40.	Typed state and authority interfaces.	Material ambiguity sometimes advanced prematurely; lexical rubrics affected some scores.
Planner	Next task 37/40; fresh-frame use 37/40; history and completion routing 40/40; grounding 22/40; capability fence 32/40.	Fresh explicit state and trusted history.	Spatial relation precision and unsupported- request fencing.
Memory	Candidate construction 40/40 and empty/undersized-store handling 40/40. Four original semantic aims stopped before inference.	Deterministic capped retrieval layer.	Thread/checkpointer defect prevented a learned semantic estimate in the original component block.

Table B.2: Selected component and integration diagnostics (continued).

Boundary	Direct observations	Strength	Important limitation
Monitor	Criteria 35/40; state 25/40; stability 30/40; progress timing 40/40; error routing 35/40.	Host timing and progress policy.	Wrong-object, slip, and unexpected states often remained ongoing.
Validator	Checklist 35/40 lexically and 40/40 in semantic audit; final state 28/40; evidence handling 25/40; reset isolation 40/40.	Checklist coverage and lifecycle isolation.	Unsupported certainty and occasional false completion/rejection.
Execution surrogate	Supported compiler calls 20/20, unsupported rejection 5/20; SAM and anchor rows 28/40; oracle-coordinate motion 35/40.	Supports a real simulated state change.	Compiler permissiveness, detection coverage, and IK are separate limitations.
Named interfaces	HRI final reporting 40/40; Planner–Monitor success route 30/40; timeout route 35/40; Planner–Validator final route 27/40.	Typed host state can contain variable learned outputs.	Memory-linked interfaces were blocked by the original Memory configuration defect.
Workflow ladder	Exact-command ceiling 39/40 assisted and 28/40 strict; raw ambiguous path 4/40; HRI level 28/40; full conjunctive level 4/40.	Exposes integration stages.	Requirements change at every level, so rates are not a causal component ranking.

B.3 Fresh-Replanning Ablations

Table B.3: Deterministic host-policy replanning ablations.

Study	Treatment	Control	Effect and boundary
AB-RPL-F	Fresh replanning recovered 40/40 stable-failure traces.	Frozen queue recovered 0/40.	Paired difference +1.00, exact McNemar $p \approx 1.82 \times 10^{-12}$. Planner outputs were deterministic fixtures.
AB-RPL-T	Automatic timeout replanning recovered 40/40 with zero operator interventions.	Attention-gated policy recovered 40/40 with 45 operator interventions.	Same recovery and safety; interaction-cost improvement at the host-policy boundary.

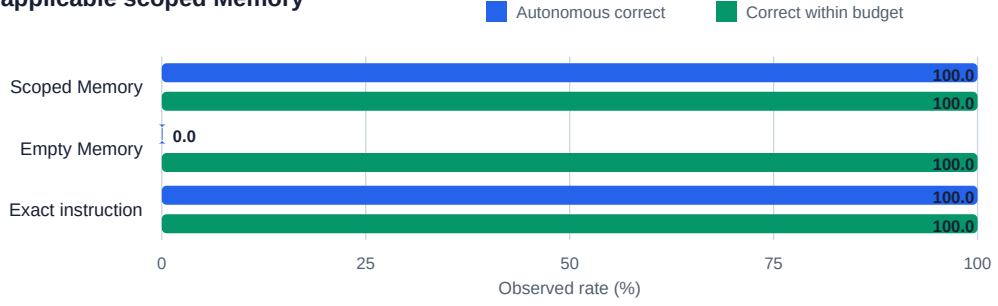
C Focused Experiment Results

C.1 Applicable Memory, Conflict, and Explicit Override

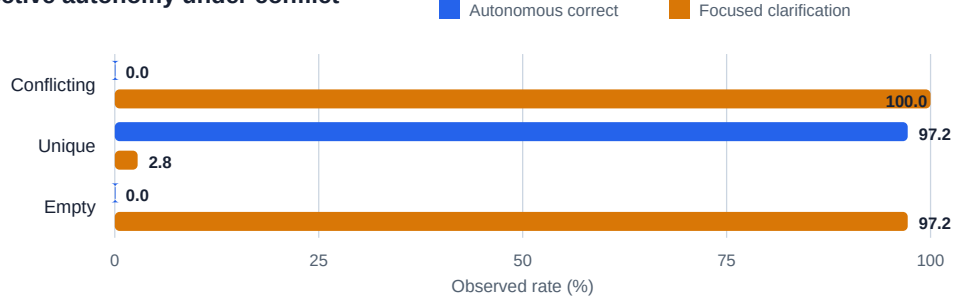
Table C.1: Condition-level results for EXP-01 through EXP-04.

Study	Condition	Primary pre-action result	Correct within budget	Boundary observation
EXP-01	Scoped Memory	Autonomous correct 36/36; mean clarification 0	36/36	Perfectly scoped retrieval 9/36; no mutation or control call
	Empty Memory	Autonomous correct 0/36; mean clarification 1	36/36	Same attainable correctness after dialogue
	Exact instruction	Autonomous correct 36/36; mean clarification 0	36/36	Descriptive contract ceiling
EXP-02	Conflicting Memory	Goal before resolution 0/36; focused clarification 36/36	36/36	Both conflicting records returned 36/36; no control call
	Unique Memory	Autonomous correct 35/36	36/36	Matching record 36/36; wrong-task exclusion 21/36
	Empty Memory	Focused clarification 35/36; autonomous correct 0/36	35/36	Uncertainty control
EXP-03	Opposing Memory	Autonomous composed 20/36; explicit board 33/36	33/36	Historical destination 0/36; clarification-observer sensitivity
	Aligned Memory	Autonomous composed 33/36; explicit board 36/36	35/36	Original measurement retained unchanged
	Empty Memory	Autonomous composed 0/36	34/36	Focused cube clarification 34/36
EXP-04	Opposing Memory	Autonomous composed 33/36; explicit board 35/36	35/36	Historical destination 0/36; wrong-task exclusion 32/36
	Aligned Memory	Autonomous composed 36/36; explicit board 36/36	36/36	Held-out replication ceiling
	Empty Memory	Autonomous composed 0/36; explicit board 36/36	36/36	Focused cube clarification 36/36

(a) EXP-01: applicable scoped Memory



(b) EXP-02: selective autonomy under conflict



(c) EXP-04: held-out explicit-override replication

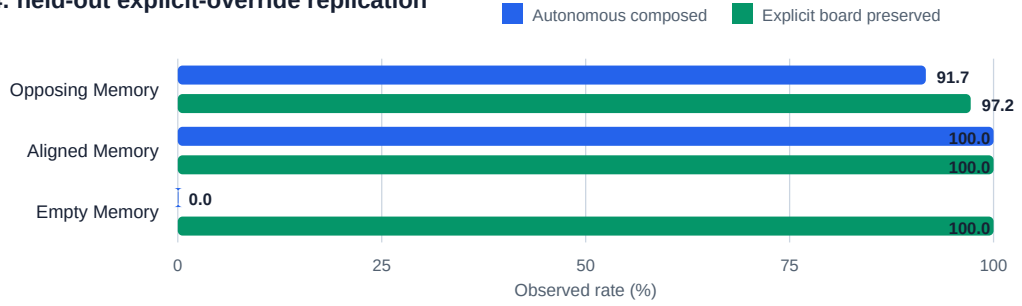


Figure C.1: Focused intent-resolution outcomes. Panel (a) distinguishes autonomy from attainable correctness; panel (b) shows selective autonomy under unique rather than conflicting evidence; panel (c) shows held-out explicit-current preservation.

C.2 Longitudinal Preference Lifecycle

Table C.2: EXP-05 lifecycle outcomes across eight independent stores.

Operation	Direct outcome	Interpretation
Create initial preference	8/8	Initial authorised persistence succeeded.
Immediate direct recall	8/8	Clean short-term retrieval ceiling.
Initial HRI use	8/8	Retrieved value formed the correct no-motion goal.
User isolation	7/8	One lifecycle crossed the registered ownership boundary.
Task isolation and abstention	7/8 each	Related task records were not perfectly isolated.
Explicit update and ID preservation	7/8 each	One update failed to retain the intended identity.
Updated-value direct recall	7/8	Stored update was usually immediately retrievable.
Updated HRI use	5/8	Correct retrieval did not always yield correct downstream staging.
Delayed direct recall	2/8	Focal identity was often lost after related mutations.
Delayed HRI use	3/8	One HRI goal was correct without the registered surviving-focal-record predicate.

Table C.2: EXP-05 lifecycle outcomes (continued).

Operation	Direct outcome	Interpretation
Revocation verified	8/8	The selected secondary record was absent after revocation, although only 2/8 revocation operations matched the intended focal lifecycle route.
Final direct recall	2/8	Robust long-term preference preservation is not supported.
Store consistency	240/240 interactions	Serialization remained valid even when semantic identity was wrong.

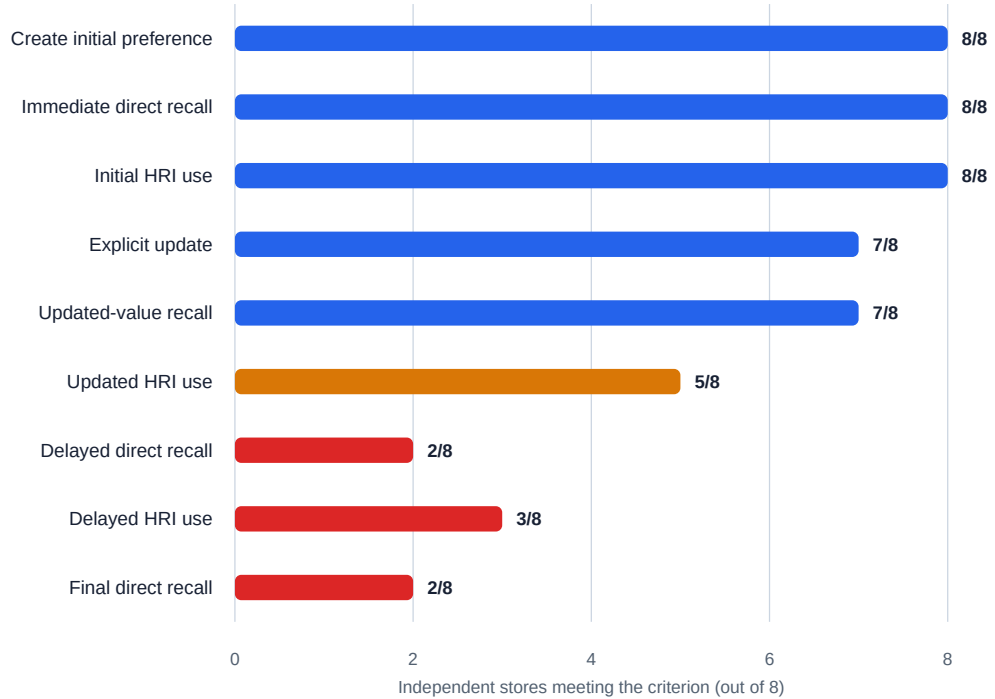


Figure C.2: EXP-05 lifecycle trajectory. Immediate creation and use were reliable, while delayed focal-record recall deteriorated after semantically related mutation pressure.

C.3 Monitor and Validator Ablations

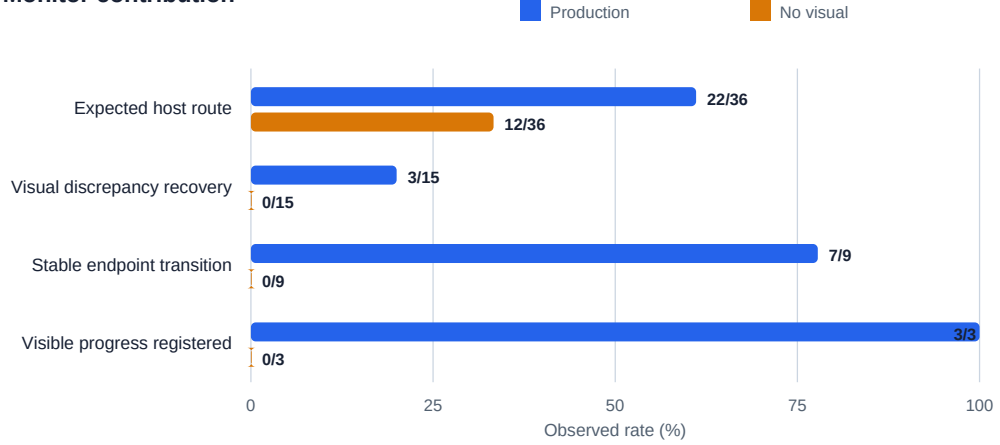
Table C.3: EXP-06 and EXP-07 paired ablation outcomes.

Study	Outcome	Production	Control	Paired effect or interpretation
EXP-06	Expected host route	22/36	12/36	+27.8 pp; cluster 95% interval +8.3 to +50.0 pp
	Visual discrepancy recovery	3/15	0/15	+20.0 pp; weak absolute recovery
	Stable endpoint transition	7/9	0/9	Clear visual endpoint contribution
	Visible progress registered	3/3	0/3	Descriptive one-cluster effect
	Safe hold / timeout	6/6 and 3/3	6/6 and 3/3	Deterministic host contribution, not learned Monitor gain
	Schema-invalid attempts	7/78	0 model calls	Retained as production capability observations
EXP-07	Expected final route	35/36	15/36	+55.6 pp; cluster 95% interval +25.0 to +83.3 pp
	False completion, incomplete	0/21	21/21	−100 pp
	Correction requested, incomplete	21/21	0/21	+100 pp

Table C.3: Feedback-gate ablations (continued).

Study	Outcome	Production	Control	Paired effect or interpretation
	Complete state finalised	14/15	15/15	−6.7 pp; one false rejection
	Exact criterion vector	34/36	15/36	Descriptive localisation of visual disagreements

(a) EXP-06: Monitor contribution



(b) EXP-07: Validator contribution and cost

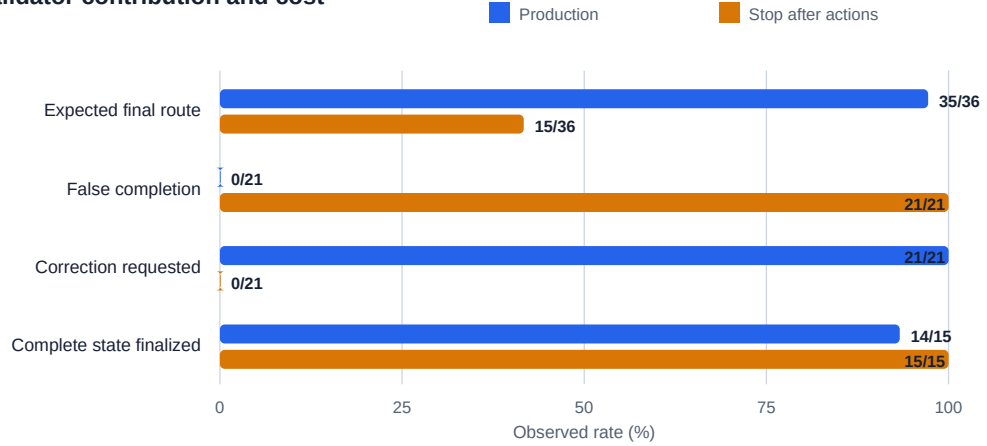


Figure C.3: Boundary-specific feedback effects. Monitor improved endpoints and progress more than discrepancy recovery; Validator prevented false completion at the cost of one controlled false rejection.

C.4 Focused Six-Family Competency Envelope

Table C.4: EXP-08 family-level outcomes.

Family	Registered event	Final outcome	Safety	Strict grounding	Oracle continuation
Slip	2/3	2/3	3/3	2/2 attempted	0/3
Timeout	2/3	2/3	3/3	1/1 attempted	1/3
Occlusion	3/3	3/3	3/3	3/3	0/3
Planned outage	3/3	3/3	3/3	3/3	0/3
Scoped preference	3/3	3/3	3/3	3/3	0/3
Explicit override	2/3	3/3	3/3	2/2 attempted	1/3

Table C.4: EXP-08 family outcomes (continued).

Family	Registered event	Final outcome	Safety	Strict grounding	Oracle continuation
Overall	15/18	15/18	18/18	14 strict completions	2/18

C.5 Focused Simulated End to End

Table C.5: EXP-09 contract-type outcomes. Every type obtained 3/3 expected Memory records, goal contracts, confirmations, correct first actions, physical goals, and safety.

Contract type	Grounding / action count	Physical / Validator	Strict chain	Terminal states
Omitted board	Strict 3/3; expected count 3/3	Physical 3/3; Validator 3/3	3/3	COMPLETE: 3
Omitted cube	Strict 3/3; expected count 0/3	Physical 3/3; Validator 0/3	0/3	NEEDS ATTENTION: 2; EXECUTING: 1
Explicit override	Strict 2/3; one assisted; expected count 2/3	Physical 3/3; Validator 3/3	2/3	COMPLETE: 3
User scoped	Strict 3/3; expected count 3/3	Physical 3/3; Validator 3/3	3/3	COMPLETE: 3
Overall	Strict 11/12; expected count 8/12	Physical 12/12; Validator 9/12	8/12	COMPLETE: 9; NEEDS ATTENTION: 2; EXECUTING: 1



Figure C.4: EXP-09 stage funnel. Contract and physical-state success remain distinct from all-strict grounding, Validator finalisation, and the exact one-action strict conjunction.

D Representative Structured Outputs and Failure Traces

Appendix Tables D.1 and D.2 reproduce two retained EXP-09 observation paths. They are examples, not additional trials. The first shows a complete omitted-board chain. The second shows a physically correct omitted-cube placement that did not receive visual finalisation and therefore repeated the action before entering attention.

Table D.1: Representative successful and false-negative EXP-09 outputs.

Recorded field	Successful omitted-board episode	Physically correct omitted-cube episode
Episode	E2E-A-01, ctx-A01, var-09	E2E-A-02, ctx-A02, var-09
User utterance	“Move red to whichever board my saved setting chooses.”	“Place the cube I routinely select in the middle of the cyan board.”
Expected Memory record	A01-memory-01	A02-memory-01
Frozen actionable goal	Move the red cube to the cyan board; preserve all other cubes.	Place the blue cube at the centre of the cyan board.
First compiler source/target	red cube / cyan board	blue cube / cyan board
Memory, goal, and confirmation	All observed	All observed
Strict grounding	True	True
Physical actions	1	3
Simulator-truth physical goal	True	True
Validator complete	True	False
Terminal state	COMPLETE	NEEDS_ATTENTION
Model calls / latency	33 / 64.16 s	98 / 138.79 s
Scientific interpretation	Complete preference-to-physical-state and evidence chain.	Intent and motion were correct, but a low-contrast blue-on-cyan visual false negative caused repetition and bounded attention.

Successful structured output (abridged)

```

memory_lookup_observed: true
expected_memory_records_observed: true
goal_contract_correct: true
confirmation_observed: true
first_source_query: red cube
first_target_query: cyan board
physical_action_count: 1
strict_grounding_completed: true
physical_goal_satisfied: true
validator_finalized_complete: true
terminal_runtime_state: COMPLETE

```

False-negative structured output (abridged)

```

memory_lookup_observed: true
expected_memory_records_observed: true
goal_contract_correct: true
confirmation_observed: true
first_source_query: blue cube
first_target_query: cyan board
physical_action_count: 3
strict_grounding_completed: true
physical_goal_satisfied: true
validator_finalized_complete: false
terminal_runtime_state: NEEDS_ATTENTION

```

Table D.2: Authority used to score the representative output fields.

Stage	Recorded evidence	Scoring authority
Preference	Lookup trace and exact returned record IDs	Frozen expected record list; no simulator state
Contract	Goal proposal, ID/revision, confirmation event	Registered semantic contract and host confirmation record
First action	Compiler source and target queries	Registered expected pick/place program
Grounding	Strict SAM/RGB-D attempt and optional assisted lineage	Strict call record; assistance never repairs the strict flag
Physical state	Settled object pose and distractor state	Hidden MuJoCo state used only after execution for scoring
Monitoring and validation	Publication-bound assessments and final report	Ordinary third-person frames plus typed host aggregation
Terminal status	Controller state, guards, and immutable history	Deterministic host state machine

E Supplementary Main-Body Tables

E.1 Related-System Positioning

Table E.1: Comparison of related personalised and agentic robot systems. “Adaptation” refers to changes beyond the immediate inference context.

System	Preference or interaction evidence	Execution feedback	Principal distinction from PrefMem
TidyBot (Wu et al. 2023)	Examples summarised into personalised placement and primitive rules.	Repeated perception and execution during cleanup.	Focuses on preference generalisation rather than consented record lifecycle, exact confirmation, or independent final validation.
LLM-Personalize (Han et al. 2025)	Demonstrations and preference-labelled self-training data.	Iterative planning from a scene graph.	Personalises the planner through optimisation rather than an explicit persistent record store.
APRICOT (Wang et al. 2025)	Visual demonstrations and active preference queries.	World-model feedback and constraint-aware plan refinement.	Learns among candidate preferences; PrefMem governs explicit persistent records, conflicts, overrides, and consent.
KnowNo (Ren et al. 2023)	Ambiguity represented as a calibrated action prediction set.	Human help is requested when the set is not a singleton.	Provides conformal calibration; PrefMem instead separates semantic, historical, and execution attention states.
Hi Robot (Shi et al. 2025)	Current prompts, observations, and mid-task user feedback.	High-level VLM repeatedly instructs a low-level VLA.	Does not centre persistent preference provenance or exact goal-contract confirmation.
Agentic Robot (Yang et al. 2025b)	High-level task instruction and visual observations.	Temporal verification, retry, and recovery for planned subgoals.	PrefMem adds consented preference memory, horizon-one fresh replanning, and a separate whole-goal Validator.
MEM (Torne et al. 2026)	Within-episode video history and semantic event summaries.	Memory conditions high- and low-level VLA policies.	Addresses embodied task memory rather than cross-interaction user preference authority.
RoboClaw (Li et al. 2026)	Structured task and working memory.	Tool-based skill monitoring, retry, recovery, and human escalation.	Unifies execution with data and policy refinement; PrefMem keeps model weights fixed and separates execution history from user preference memory.
PrefMem	Explicitly mutable, scoped preference records plus current user input.	One published task, temporally fenced Monitor evidence, fresh replanning, and independent final validation.	Studies preference provenance, exact goal authorisation, deterministic state ownership, and evidence-gated terminal decisions as one inspectable orchestration boundary.

E.2 Architectural Alternatives

Table E.2: Architectural trade-off between a monolithic tool-using agent and PrefMem’s role-specialised graph.

Criterion	Monolithic agent	Role-specialised PrefMem
Model residency	One loaded model	One shared loaded model; logical roles do not duplicate weights
Prompt and tools	Union of all histories, instructions, schemas, and tools	Role-specific context and least-authority tool surfaces
State ownership	Primarily model/conversation mediated	Deterministic runtime and controller own authoritative state
Temporal behaviour	One loop multiplexes dialogue, observation, and control	Separate user-turn, planning, monitoring, validation, and execution loops
Verification	Planner can influence or certify its own result	Separate Monitor and Validator evidence routes; host derives completion
Testability	End-to-end behaviour is simpler but failures are opaque	Components, interfaces, and host policies can be tested separately
Cost	Fewer calls and lower orchestration overhead	More calls, schemas, latency, and integration complexity

E.3 Interaction Contracts

Table E.3: Principal interaction contracts and authority boundaries.

Source	Destination	Artifact	Authority rule
User	HRI	Request, clarification, confirmation, guidance	User authorises ambiguous semantics and persistence
HRI	Memory	Labelled retrieve or mutate request	Memory can return or change only request-valid records
HRI	Runtime	Preview, confirm, resume, or replan tool call	Runtime validates state and exact identifiers
Runtime	Planner	Preview payload or PlannerCycleRequest	Planner proposes; host assigns IDs and selects task one
Runtime	Monitor	STEP PublishedTask	Monitor observes only the current step
Runtime	Validator	Goal for compilation or final publication	Validator cannot rewrite goal or host-derived status
Runtime	Executor	STEP PublishedTask	Executor changes the scene but cannot establish completion
Monitor / Validator	Runtime / Controller	Typed assessment or error	Identity, time, phase, and criteria must match
Controller	Planner	Immutable ExecutionRecord on next cycle	Only host code appends trusted history
Runtime	HRI	State summary, attention cause, validation report	HRI reports but cannot fabricate recovery authority

E.4 Requirement-to-Evidence Traceability

Table E.4: Requirement-to-mechanism and evidence traceability.

Req.	Primary mechanism	Principal evidence	Important boundary
R1	HRI ambiguity policy, scoped Memory use, focused clarification	C-HRI-AMB, interfaces, EXP-01 and EXP-02	Selected tabletop field resolution, not open-world intent understanding

Continued on next page

Table E.4: Requirement-to-mechanism and evidence traceability (continued).

Req.	Primary mechanism	Principal evidence	Important boundary
R2	Retrieval provenance, request-scoped mutation guards, explicit-current priority	C-MEM, EXP-02 through EXP-05	Free-text identity and scope remain longitudinal limitations
R3	Host goal ID/revision and exact confirmation tool	HRI tool and state tests, EXP-01–04 and EXP-09	Confirmation is per run and is not memory consent
R4	Stateless Planner, fresh frames, candidate-one selection, terminal history	C-PLN, workflow ladder, AB-RPL-F/T	Host ablations do not establish learned Planner quality
R5	Publication IDs, generations, phase and frame-sequence checks	C-MON-ERR, C-VAL-RESET, interaction tests	Fencing does not correct a semantically wrong but schema-valid assessment
R6	Repeated success/failure evidence and stability duration	C-MON-STABLE, EXP-06	Visual label quality limits what host aggregation can stabilise
R7	Frozen ValidationContract and host-derived broad status	C-VAL, EXP-07 and EXP-09	Prevents false completion but may cause false rejection
R8	Attention state, timeout records, repetition and cycle guards	PF-REPLAN/SAFETY, AB-RPL-T, EXP-08	Software recovery policy, not hardware safety certification
R9	Narrow PublishedTask boundary and human/MuJoCo adapters	C-EXE, calibration, EXP-08 and EXP-09	Current adapter is not a learned VLA
R10	Typed contracts, immutable attempts, traces, manifests, direct staged metrics	All suite layers and focused reports	Original suite remains development/non-confirmatory evidence

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